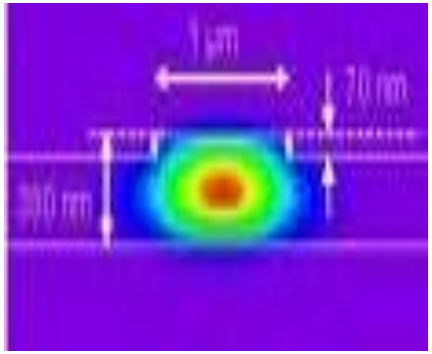
PHOTONIC DEVICES

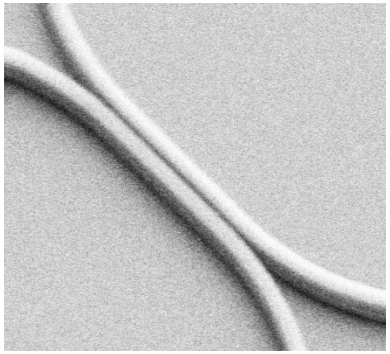
Andrea Melloni

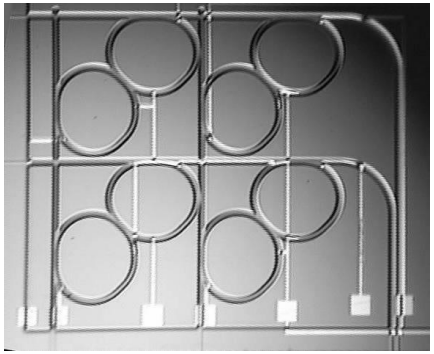
andrea.melloni@polimi.it

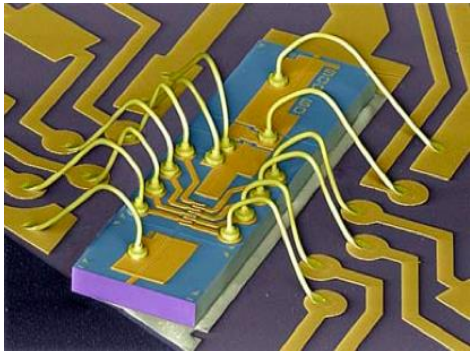




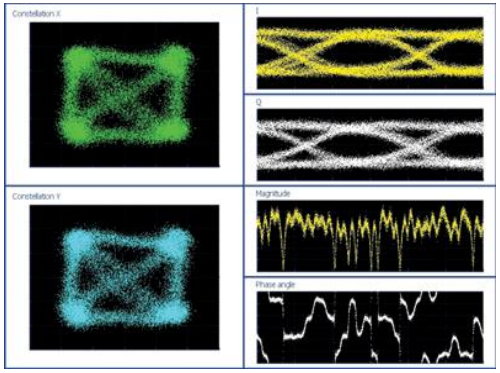


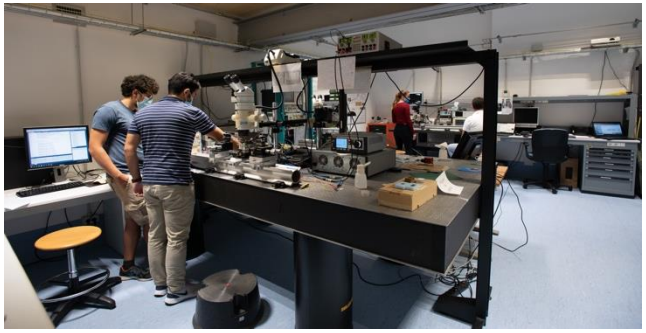


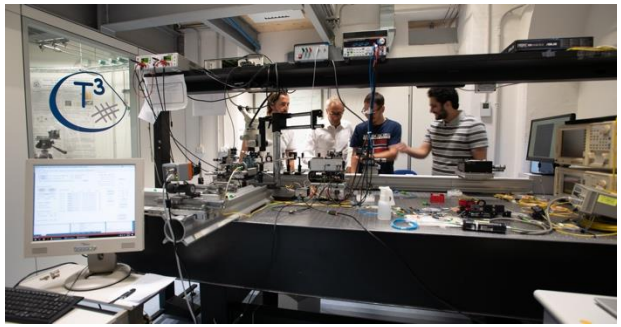














Via Colombo, 81

http://photonics.deib.polimi.it

Photonic Devices Lab

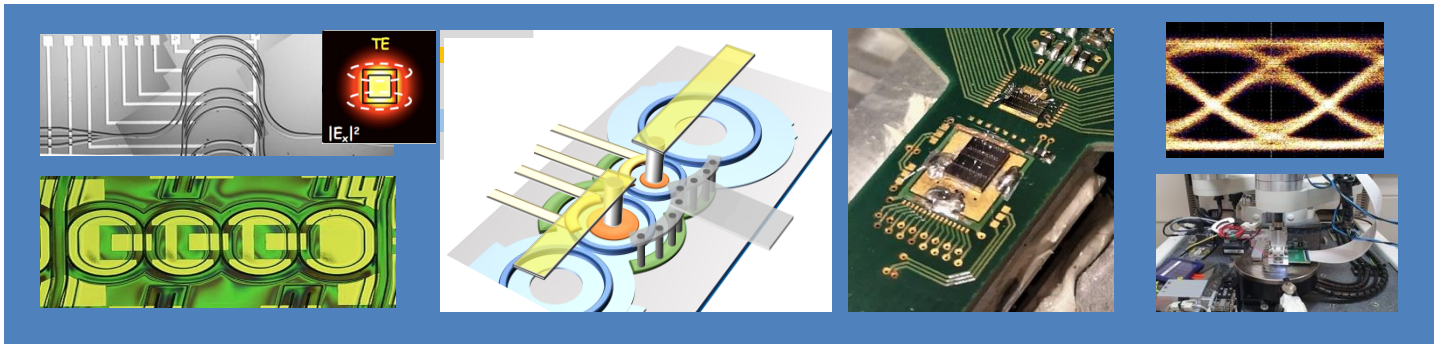
Andrea Melloni

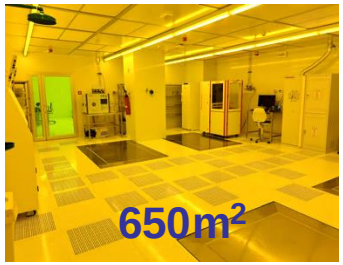
Francesco Morichetti

POLITECNICO DI MILANO

polifab POLITECNICO DI MILANO

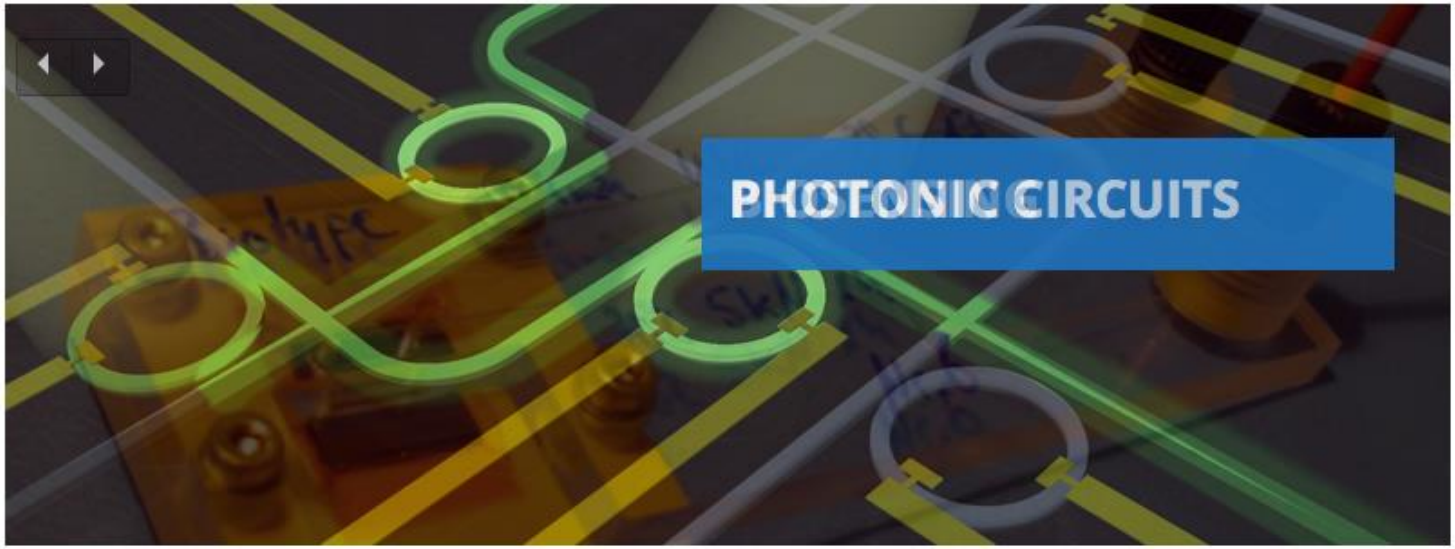
PoliMi micro- nano-technology center





Photonic Devices Group

HOME | TEAM » | RESEARCH » | PROJECTS » | PUBLICATIONS | TEACHING » | NEWS | CONTACTS



ABOUT US

The Photonic Devices Group operates at the Dipartimento di Elettronica, Informazione e Bioingegneria of the Politecnico di Milano since 1996. Our prime competences are in passive and nonlinear integrated optical circuits, from electromagnetic analysis to circuit design, from waveguides characterization to devices testing, from devices for high bit rates optical communication systems to sensors and optical interconnect. We have a world leading expertise ranging from silica-on-silicon technology, high index contrast silicon oxynitride (SiON), silicon nitride, silicon (SOI), chalcogenide and Indium Phosphide. The Labs include 5 polarization controlled setups for characterization of waveguides and components in both frequency (attenuation, insertion loss, birefringence, group delay, near field, backscatter...) and time domain (BER up to 10 Gbit/s, single pulse down to 1 ps). We own also a phase sensitive Optical Low Coherence Interferometer for tomography, inspection and advanced characterizations.

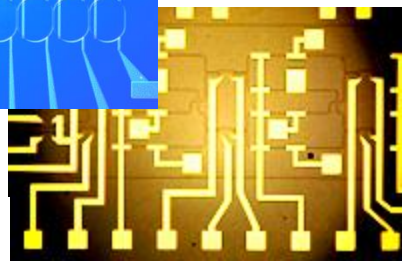
http://photonics.deib.polimi.it

>25 years of Multidisciplinary photonics@ PoliMi

POLITECNICO DI MILANO

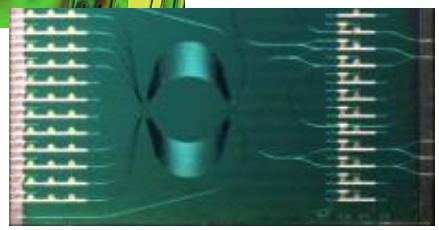
Silicon photonics

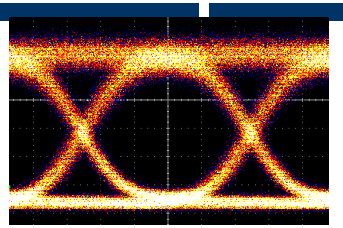


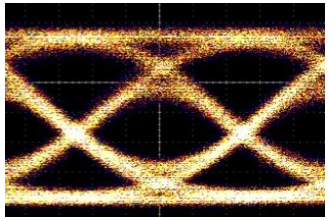




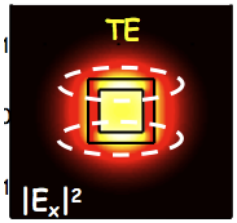
Indium phosphide

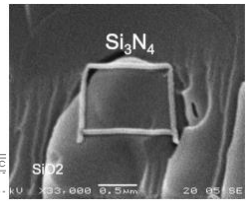


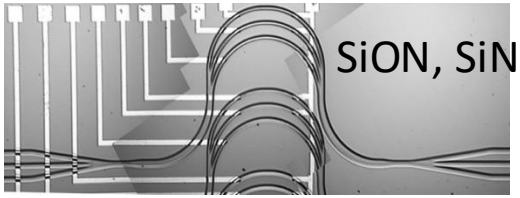




WDM up to 400 Gbit/s

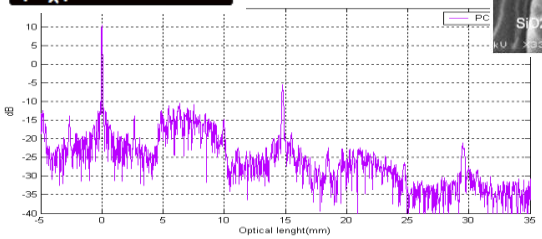




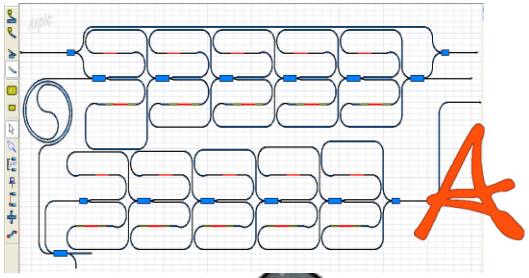


SiON, SiN

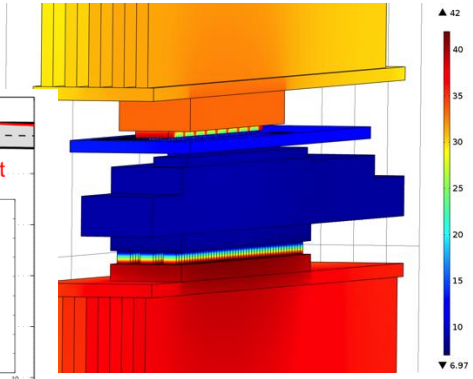
Commercial Circuit simulator

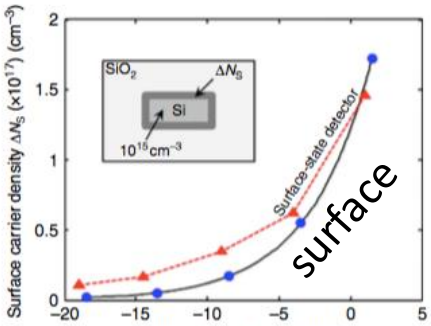


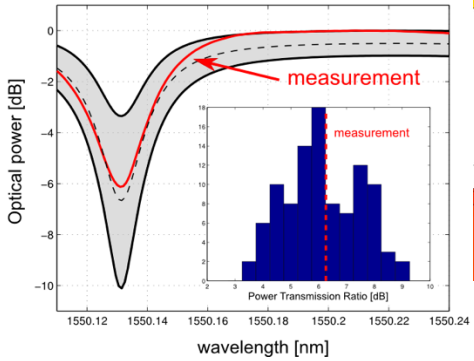
Thermal & Stress analysis



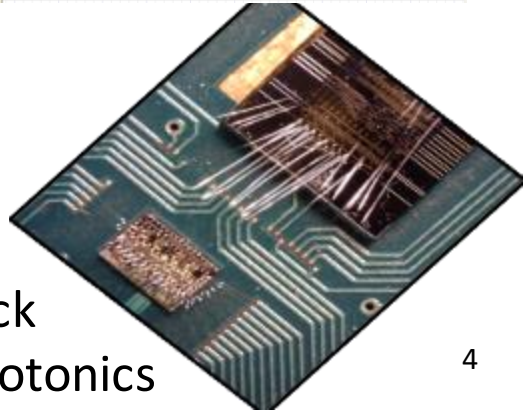
Characterization techniques





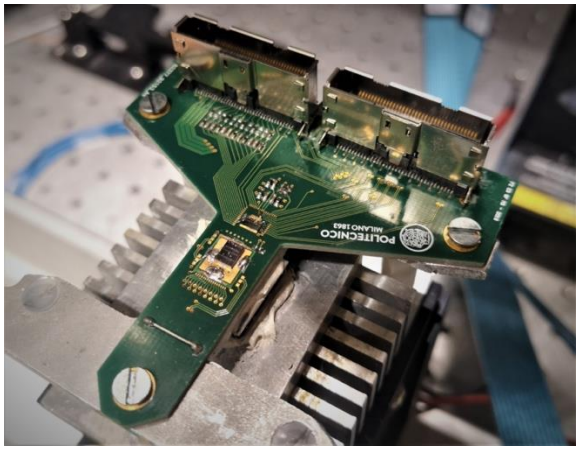


Statistical analysis



Feedback controlled photonics

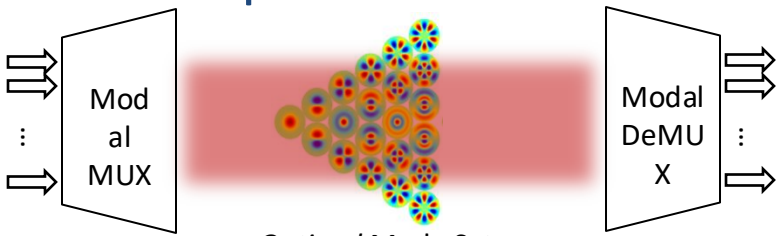
Applications

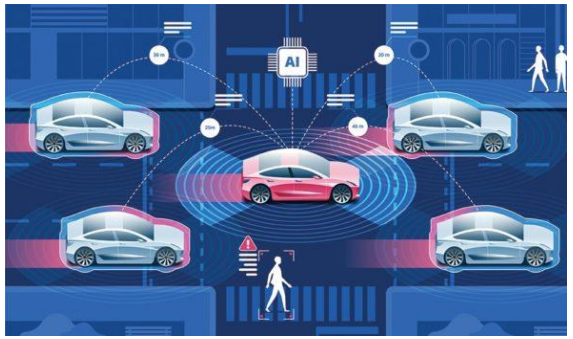




Free Space Optics

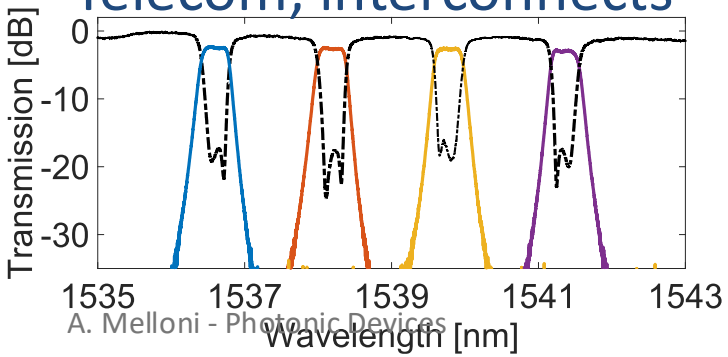
Optical MIMO

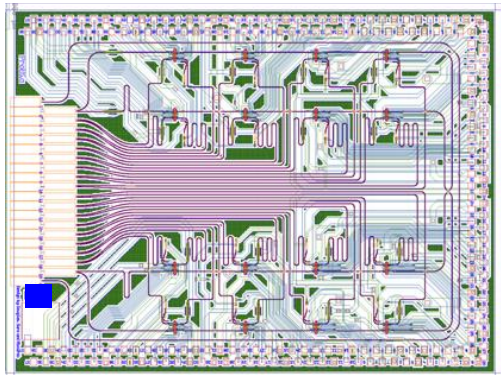




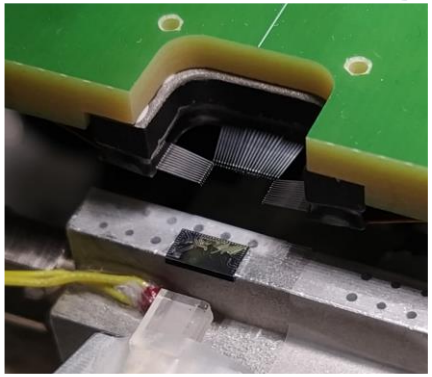
ICAS & LIDAR

Telecom, interconnects

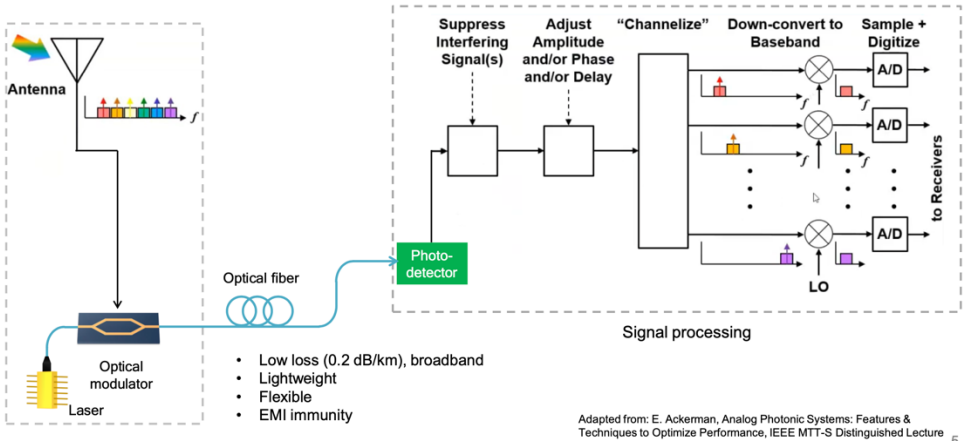


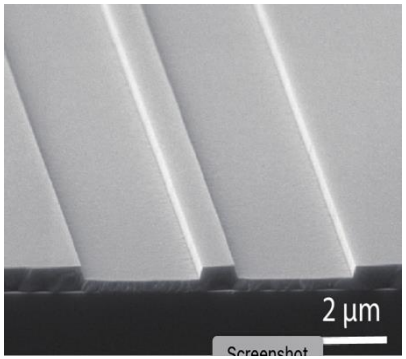


Testing



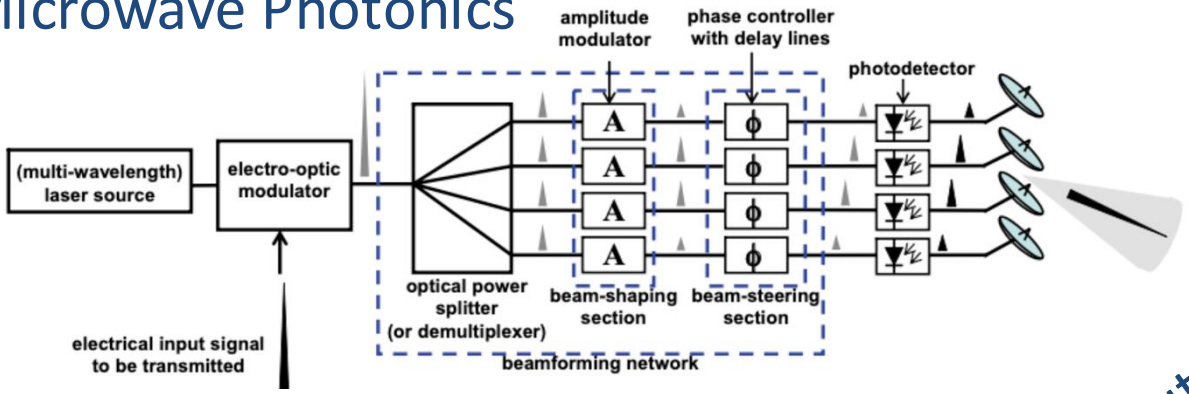
Applications

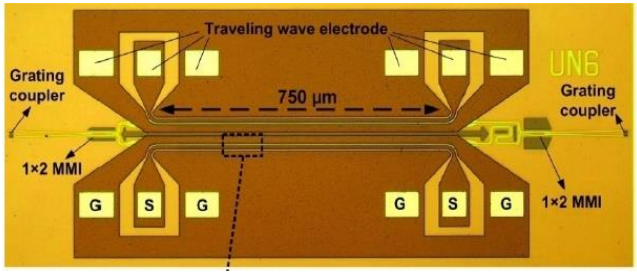


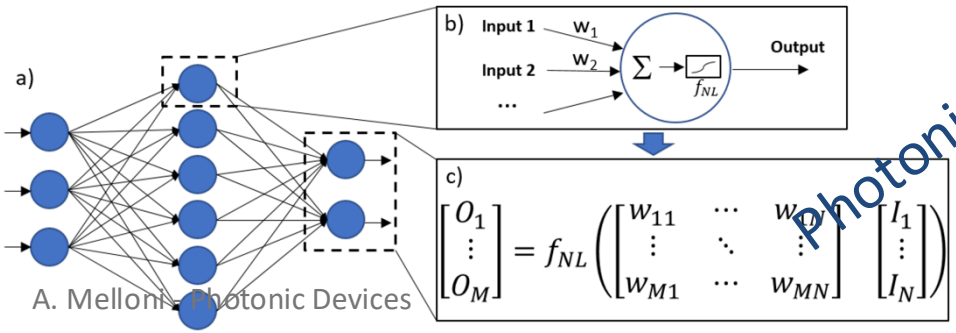


Technologies and Devices

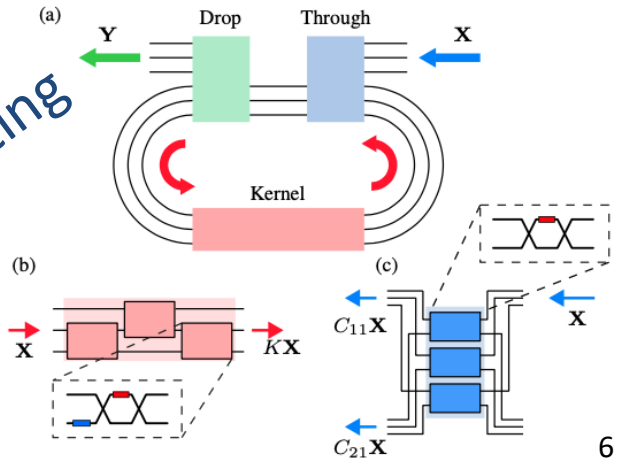
Microwave Photonics



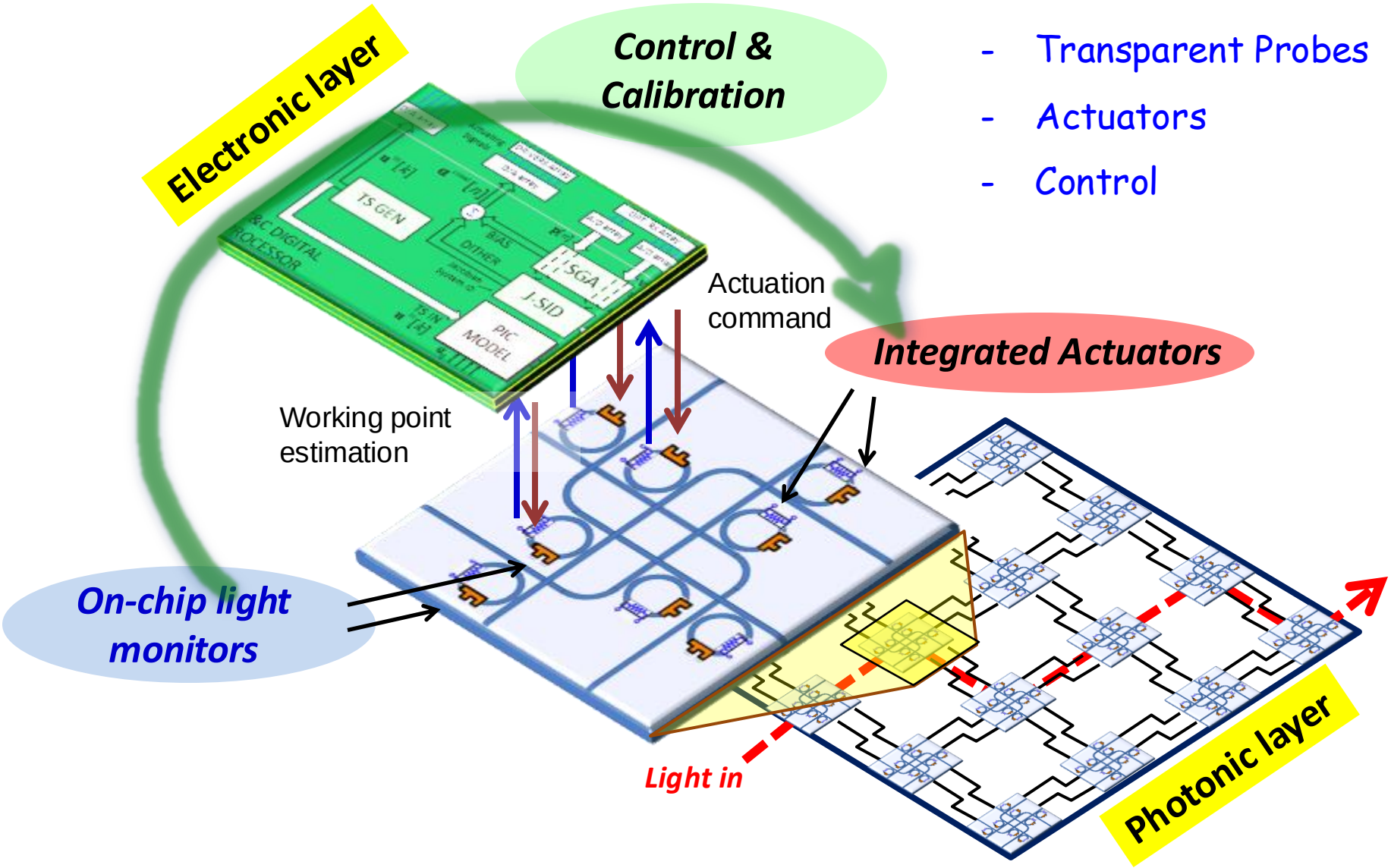




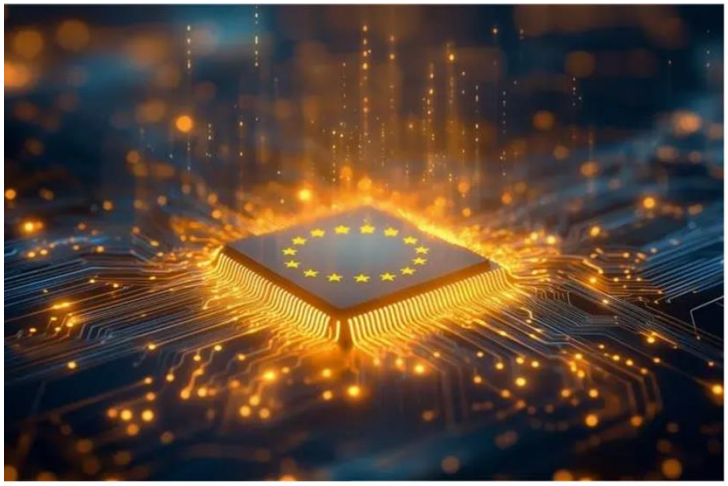
Photonic Computing



Control layer for photonic integrated circuits



PIXEurope Pilot Line in Photonics



The European Commission and Chips JU select the **PIXEurope** consortium to lead the European Pilot Line on Advanced Photonic Integrated Circuits

Coordinated by ICFO from Barcelona, PIXEurope will mobilize investments of about 400 million euros to offer unique technological capabilities to industry, with the aim of enhancing its capacity regarding photonic chips and positioning Europe as a global leader.

PIXEurope includes partners from Austria, Belgium, Finland, France, Ireland, **Italy**, Poland, Portugal, Spain, the Netherlands and the United Kingdom.

Polifab and Photonic Device group as **Competence center i**n Italy

Tasks: SiC platform, control layer, testing

Starts 1st May 2025
5 years project
Recruiting now !

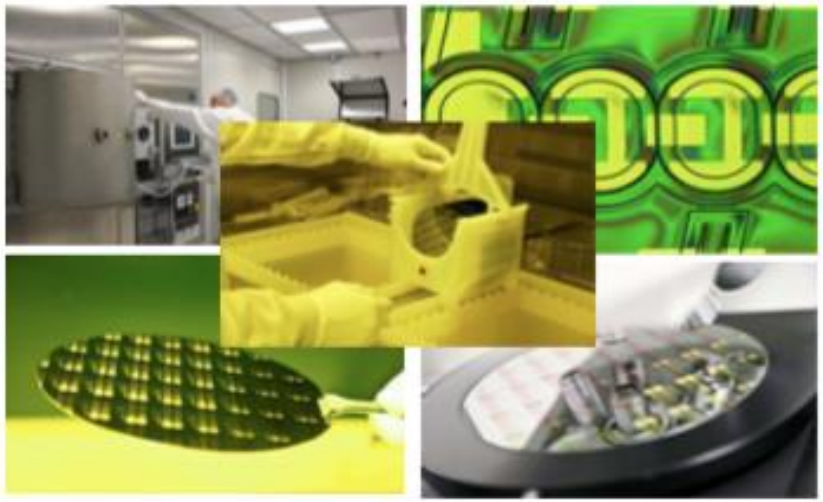
Since 2015

polifab
POLITECNICO DI MILANO



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Polifab is the open-access, micro and nanofabrication facility of Politecnico di Milano

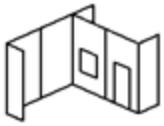


The mission of Polifab is to support and boost the activities of the research groups of Politecnico, as well as of external research institutions and industries.

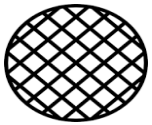
Aggregation center for collaborations between the University and industries, providing high-technological know-how for industrial applications.

Download the Annual Report ! https://www.polifab.polimi.it/reports/

Polifab at glance



+600 mq cleanroom, ISO 06



Up to 200 mm wafers



11 staff members

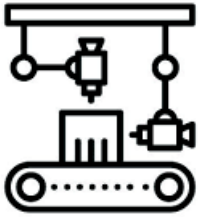


Open access



+150 researchers in 2023





~ 25 M€ tools installed

~ 40% from ST, JRC MEMS and STEAM

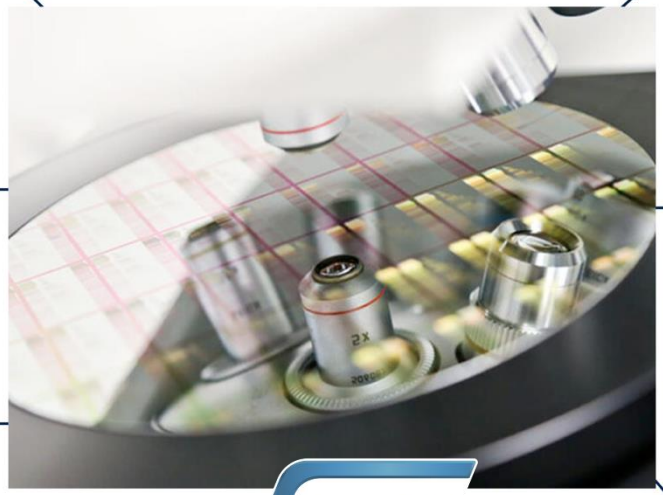
Multidisciplinary approach, flexibility

Polifab Capabilities

Wet etch and surface preparation: wafer cleanings, lift-off, wet benches for etching processes

PVD of dielectrics, metals, magnetic and Piezoelectric materials : TiO_2, SiO_2, Ti, Pt, Al, Ru, KNN, PZT

Dry etching:
Reactive Ion Etching of silicon, dielectrics and metals layers; photoresist ashing and metal Ion Beam Etching



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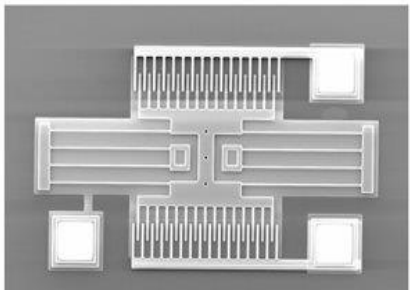
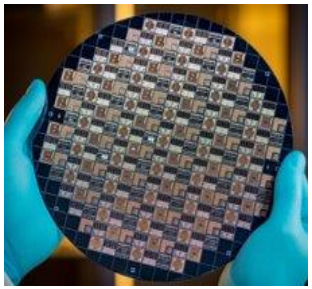
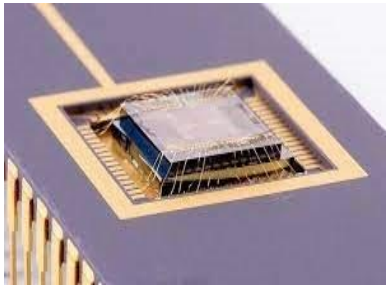
Metrology: ellipsometry, SEM/EDX, AFM, metal sheet resistance, XRD, IV/CV probe stations

Thin Films Deposition:
Thermal and e-beam evaporation, PECVD of silicon oxide, silicon nitride, MBE, PLD, ALD, direct printing

Lithography: EBL, two-photon, laser writer, maskless aligner

Several tools can offer 8" wafers processing such as PECVD, RIE, DSiE, HF release, etc.



Polifab Clean Room – 620 m²

Lithography
- Photolithography
- Maskless photolitho
- Electron beam lithography
- Thermal scanning lithography

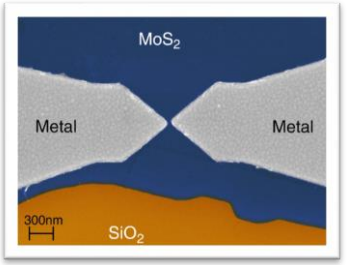
Deposition
- Electron beam evaporation
- Thermal evaporation
- Sputtering
- Chemical vapor deposition
- Electrodeposition

Etching
- Reactive ion etching
- Ion beam etching
- Plasma ashing
- Wet etching

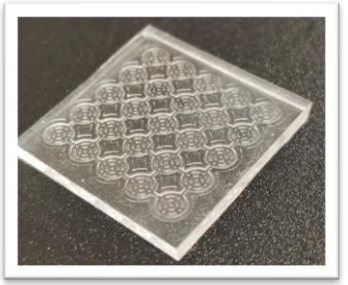
Characterization
- Optical microscopy
- Stylus profilometry
- Optical profilometry
- SEM, AFM
- Ellipsometry
- Electrical measurements
- X-ray diffraction
- EDX

Backend
- Annealing ovens
- Rapid thermal annealing
- Wire bonding
- Dicing
- Pick & Place
- Alignements tools

FROM NANO TO MILLI



D. Ielmini, DEIB



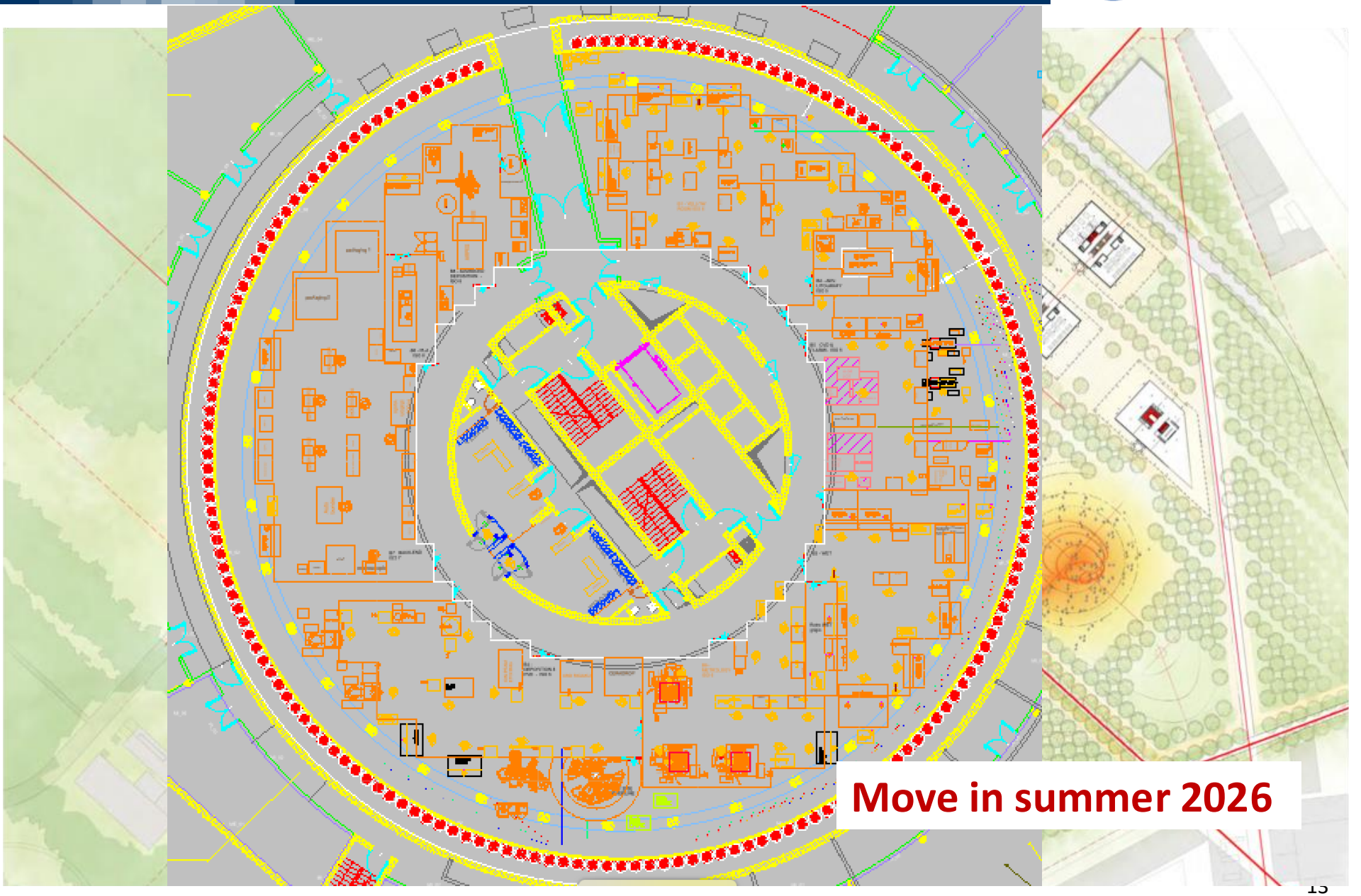
E. Bianchi, DCMIC

Available in near future:

- Two-photon polymerization tool
- Sputtering for metals and oxides
- Electron beam lithography
- Micro-Raman spectroscopy
- ……..

http://www.polifab.polimi.it

The future: Gasometro 2 – Bovisa Campus

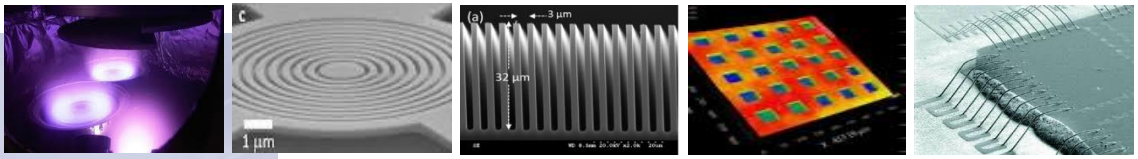


Move in summer 2026

Polifab – Tomorrow (2026)

+1200 mq: the largest university cleanroom in Italy

Deposition | Lithography | Etching | Characterization | Backend





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PoliHub

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LUXOTTICA

POLITECNICO
MILANO 1863

Gasometro 2 – Innovation Hub, a whole ecosystem for R&D

Polifab as an aggregation center for technologies

Bridge the gap between fundamental academic research and industry

AGGREGATION IS STRENGHT

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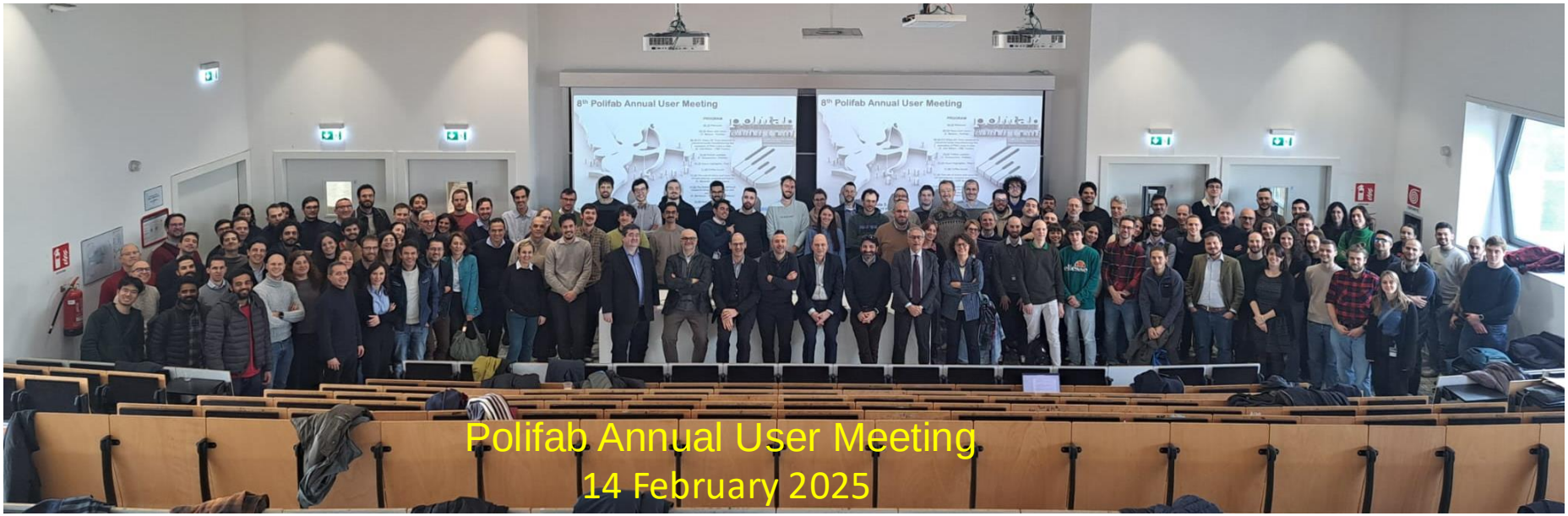
life.augmented

JRC MEMS
JRC STEAM

POLITECNICO MILANO 1863
JLab
HUAWEI

mmWave Lab

LUXOTTICA



Polifab Annual User Meeting
14 February 2025

SCUOLA:
Scuola di Ingegneria Industriale e dell'Informazione

Via Giuseppe Colombo 81
20133 Milano Italia

https://www.polifab.polimi.it

Photonics Devices (10cr)

The course in the TLC, ELN, (FIS) study plans

Program of the course

Detailed course contents

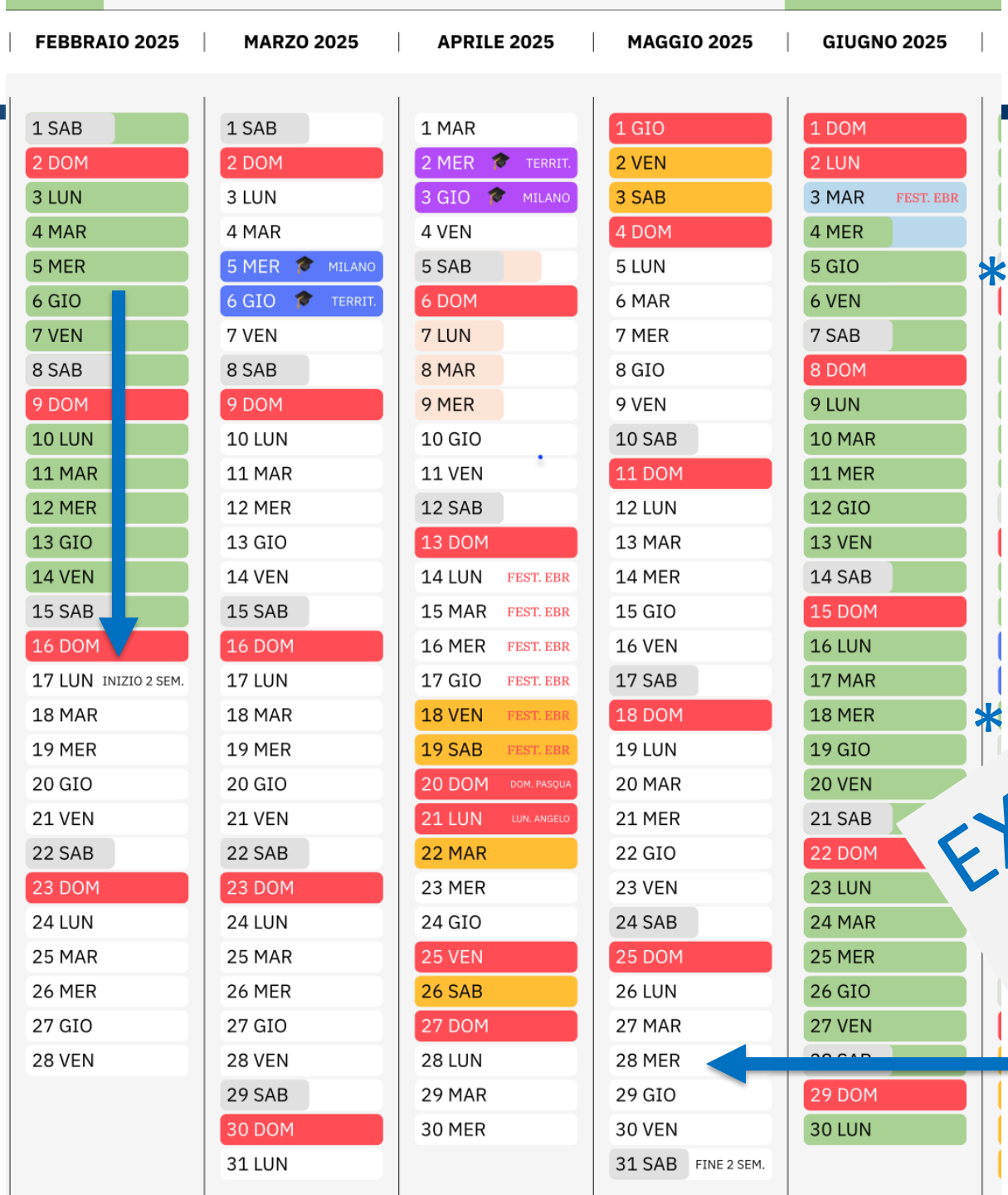
Time schedule, Exams, ...

Photonics at glance (Photonics21)

POLITECNICO DI MILANO

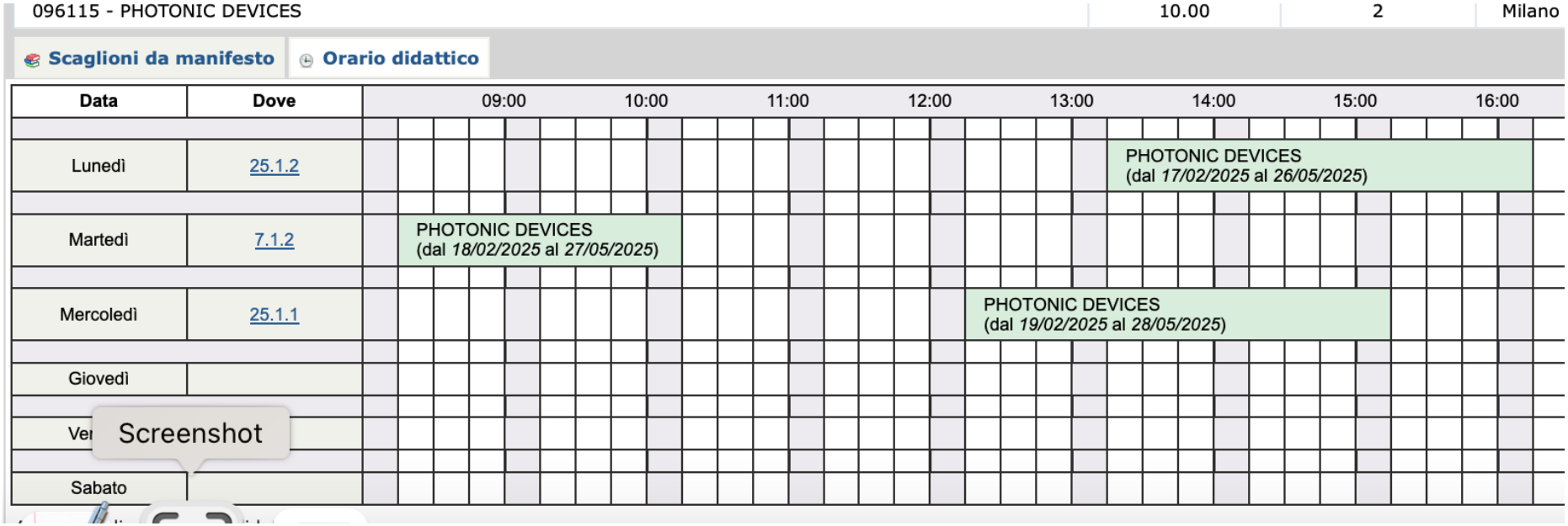
2024-25

14 weeks (too much...)



EXAMS every 2 weeks

AA 2024-2025



096115 - PHOTONIC DEVICES | 10.00 | 2 | Milano

Scaglioni da manifesto | Orario didattico

Data	Dove	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00
Lunedì	25.1.2						PHOTONIC DEVICES (dal *17/02/2025* al *26/05/2025*)		
Martedì	7.1.2	PHOTONIC DEVICES (dal *18/02/2025* al *27/05/2025*)							
Mercoledì	25.1.1					PHOTONIC DEVICES (dal *19/02/2025* al *28/05/2025*)			
Giovedì									
Ven									
Sabato									

Screenshot

Photonic Devices

- from now (17th February) to 28th May, ~14 weeks
- Organization: Frontal lessons, home works, seminars, tasks, tests (Socrative), lab visit, clean room visit, virtual experiments, and more…
- In presence (registrations available, no streaming)
- Exams:
  - Every ≈2 weeks from 5th June to 25th July, 2 dates in September
  - No exams out of sessions
  - Max 12 students per date, schedule by students via Doodle
  - Changes and cancellations are possible within 24h from exam
  - Oral discussion 20 min. 1 topic at student choice, 2 topics prof.
  - Tasks (home work) are a “plus” as well as presence

Photonic Computing

NEW !

- New course ! (5 cr)
- 2nd semester
- Prof. F. Morichetti (and Melloni)

Photonic Devices - Program

The perspective of photonics integration. Integrated optic for optical communications, sensors and optical interconnects (electronics). The generic foundry scheme in photonics. Technological platforms: glass on silicon, Lithium Niobate, Indium Phosphide, Silicon Photonics. The ubiquitous market of photonics devices. Potentialities and limits.

Optical components: Integrated-optic components. The propagation in planar guided-wave structures. Guided, radiative and leaky modes. The coupled mode theory. Bent waveguides, couplers and Y. Star couplers. Filters, (de)multiplexer and AWG. Switch. Integrated-optic modulators: phase and amplitude, novem modulation formats. Travelling wave electrodes. Fiber-optic grating: uniform, apodized and chirped. Fiber Bragg gratings. Isolator and circulator. Integrated Lasers and Phototodetectors.
Materials and technologies: Materials for integrated optic circuits. Electro-optical and magneto optical effects. Technologies for integrated optic circuits: glass on silicon, Lithium Niobate, Indium Phosphide and Silicon Photonics. The photorefractive effect and fiber Bragg gratings. Technological processes for the realization of passive components. Layer deposition, photolitography and etching. The packaging issue.

Optical circuits: The circuits for the photonic networks and switching. Components for Wavelength Division Networks. The wavelength routers. Switching fabric for optical signals. Add-drop and cross connects. Principles of optical signal processing. Components and circuits for EDFA and SOA. Gain and noise figure. SOA for optical signal processing. Optical time-domain reflectometry. Optical spectrum analyzer. Optical low-coherence interferometry.

Applications. The interconnect problem. Analysis of the electrical interconnect and the optical alternative. Limits, advantages, constraints. From short range rack to rack to intra-chip links. Link performance (comparison). Integration technologies for combining optoelectronics, optics, and electronic integrated circuits: Silicon Photonics, Indium phosfide and Polymers. Monolithic vs hybrid solutions. CMOS compatible solutions. Examples of PICs applications are considered and discussed (May). Photonic computing.

PHOTONIC DEVICES - Time schedule

Week 1
Scenario
Applications
Technologies

Week 2
Slab. Waveguides, propagation, modes, mode solver, BPM examples, type of waveguides, bent waveguides.
Birefringence, neff, ng.
Advanced aspects of waveguiding: role of Δn, backscatter, radiation, xtalk….

Week 3
Coupled mode theory, directional coupler, design of, WDM coupler, Y branches, BPM examples. MMI (star coupler later)

Week 4
Filters FIR & IIR. Mach-Zehnder, Ring resonator (filter and phase shifter, enhancement concept), AWG. Analysis and Design, Aspic examples. Advanced filters and mux (cascaded).

Week 5
Applications of passive: filters, delay lines, xconnects, OADM, software defined photonics….
Tecnologies, Glass on silicon, InP, SiliconPhotonics. Overview and differences, technological steps, fiber-waveguide coupling (butt coupling & gratings). Packaging.

Week 6
Bragg gratings, Isolator, circulator, reciprocity (time and nonlinear)
Characterizations and testing: OLCI, OTDR, reflectometry.

PHOTONIC DEVICES – Time schedule

Week 7
Modulators, generalities. LiNbO3 modulators (electrooptics, technology, MZ, electrodes, chirp, multilevel…). InP modulator and SOI modulators. Electrodes, RF frequency constraints

Week 8
Active functionalities: InP: SOA (principle, use, parameters), PD, laser (Integration of, principles, state of the art), technology. Transmitters and receivers

Week 9
The optical interconnect problem, electrical vs optical, system evaluation. Comb. Limits.
The photonic foundry scenario: PDK, circuit simulators, fabless scheme.
Photonics integration: what, why, how. Xtalk, Photonic-electronic co-integration….

Week 10-11
Silicon photonics
CMOS compatibility
Photonic computing

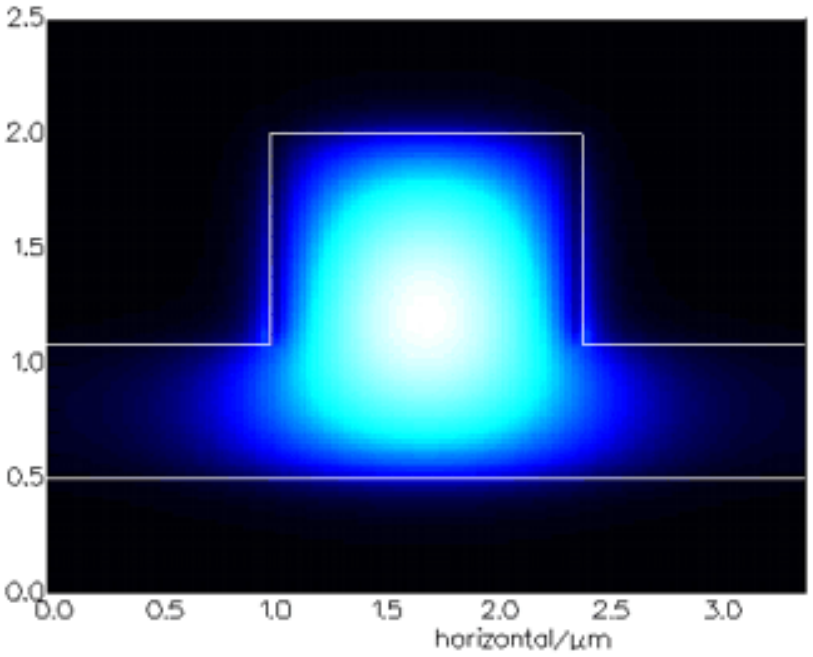
Week 12
What's hot in photonics (microwave photonics, control layer, new materials, biosensing, plasmonic, THZ, Nanoantenna, Optomechanic, photon,….) Vision, future
The roadmap towards 2020
Seminars and Visits (Photonic Devices Lab and Polifab)

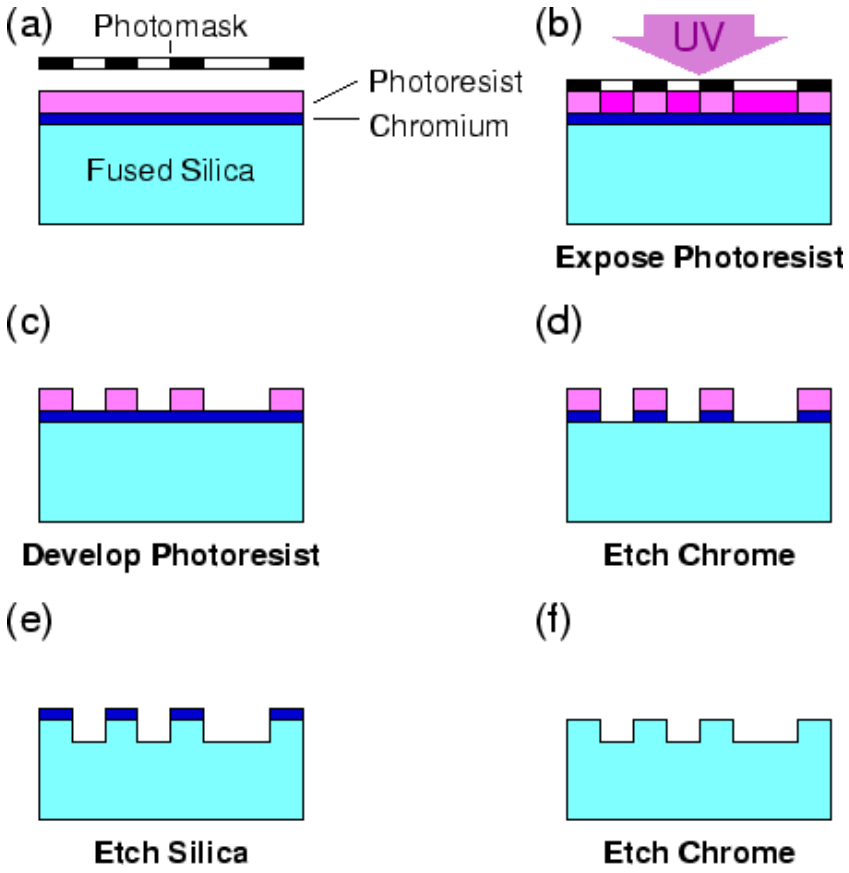
Textbook and Software
Material in Beep course webpage

Dispense ITA
Notes in ENG

T. Tamir, Guided wave optoelectronics, 1990
C-L. Chen, FOUNDATIONS FOR GUIDED-WAVE OPTICS, Wiley, 2007
C. Madsen, J. Zhao, Optical Filter Design, Wiley 2007

From the foundamentals …

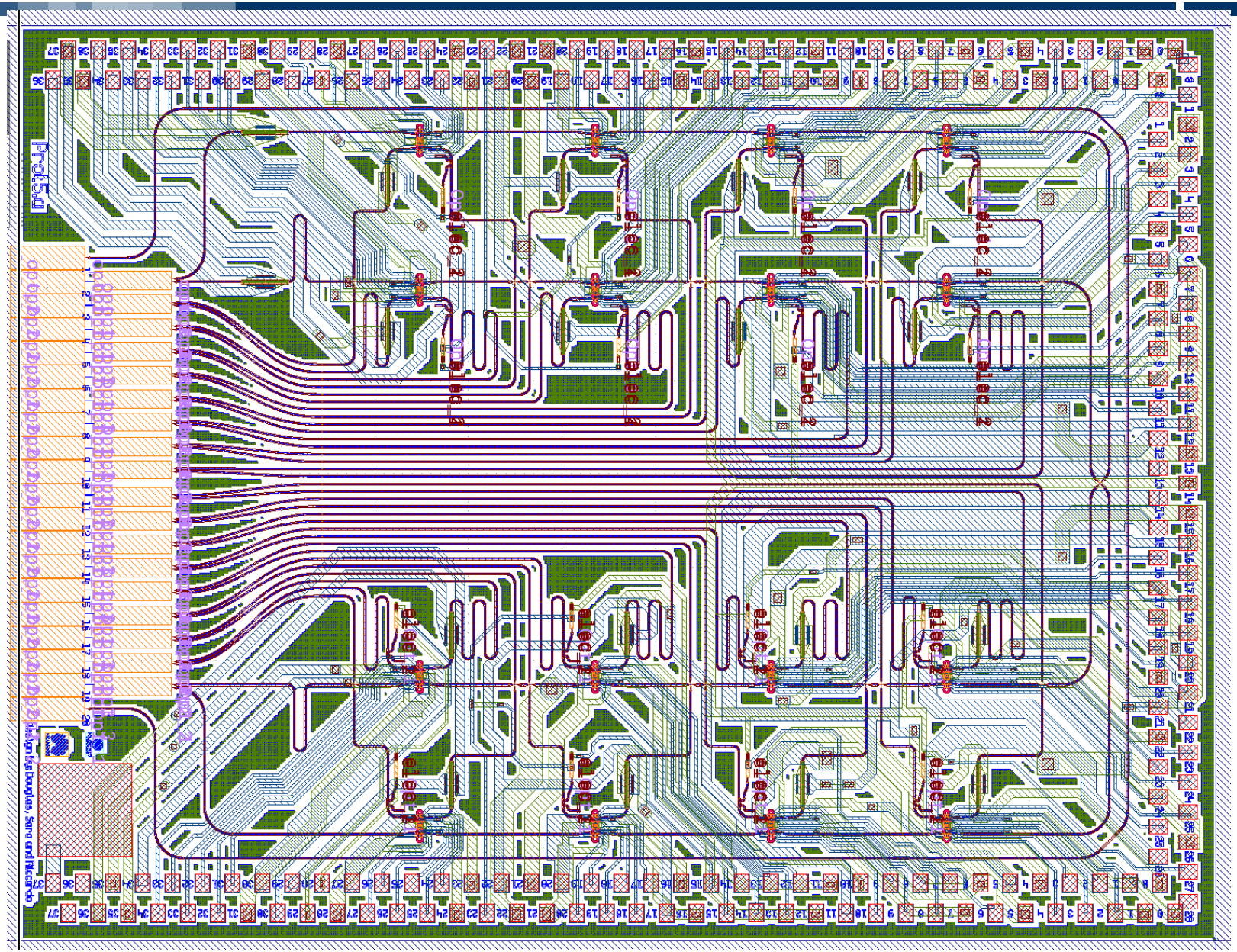




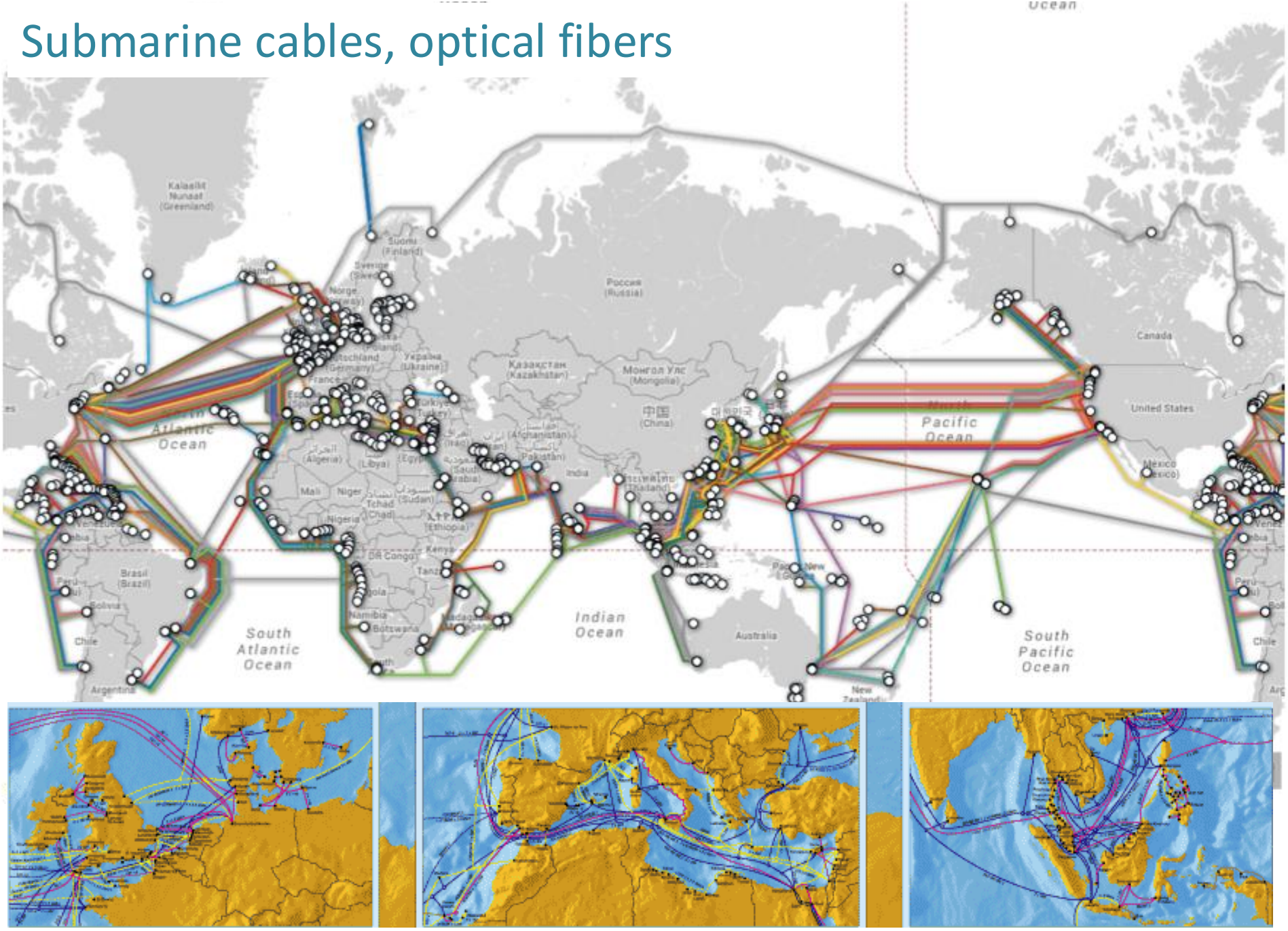
Electromagnetism…

Technology and materials

… to the state of the art of complex devices



Submarine cables, optical fibers

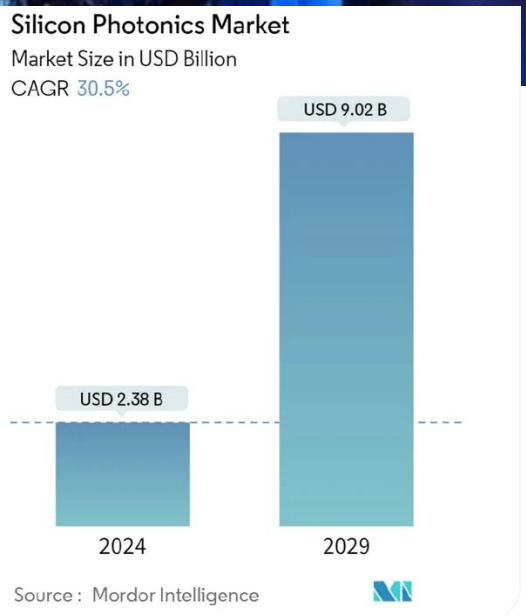


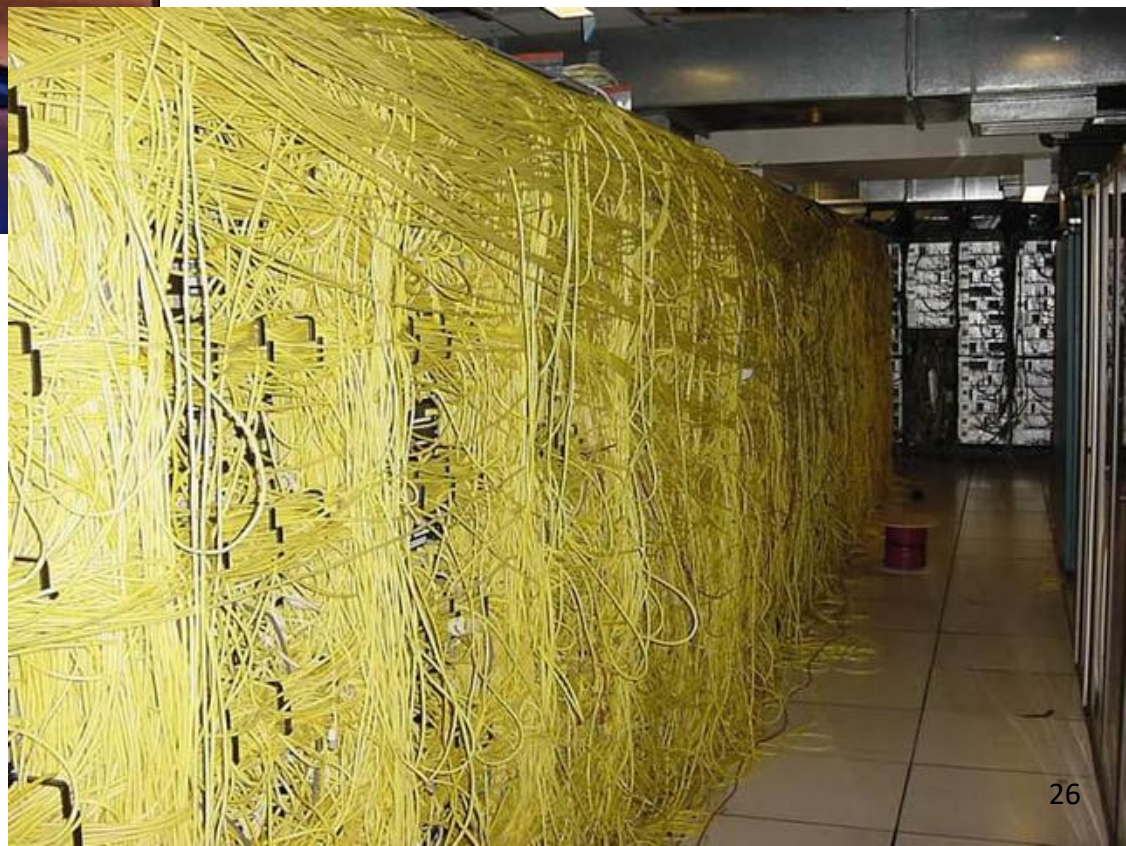
Optical Interconnects

Cisco reports total bit rate for internet traffic 2018 ~ 280 Tb/s
2023 ~ 1000 Tb/s
+30%/year



Datacenters....





HPC wire

Since 1987 - Covering the Fastest Computers in the World and the People Who Run Them

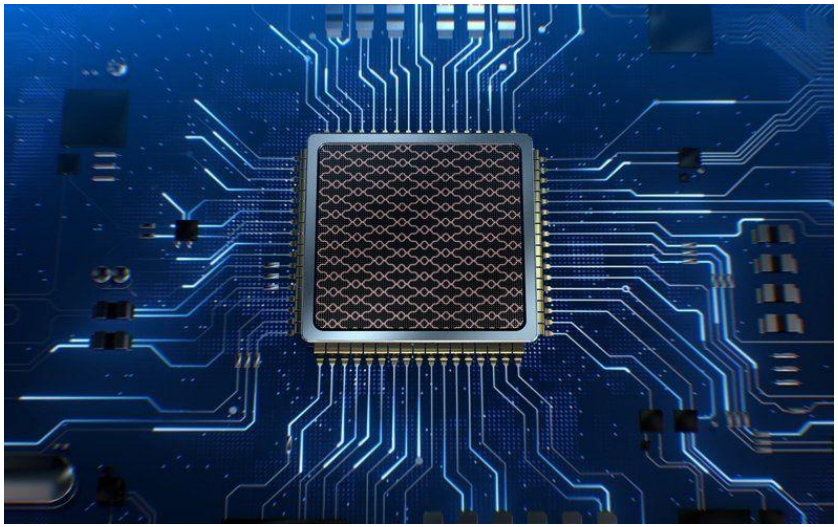
- Home
- Technologies
- Sectors

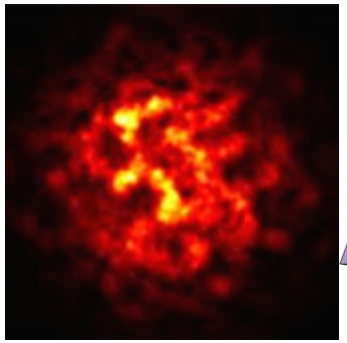
Researchers Develop Novel Analog Processor for High Performance Computing

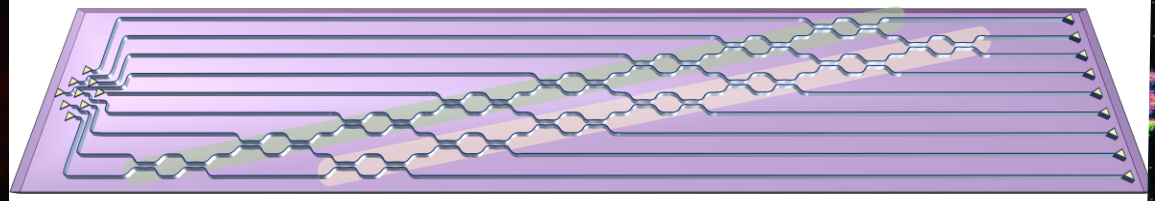
August 30, 2021

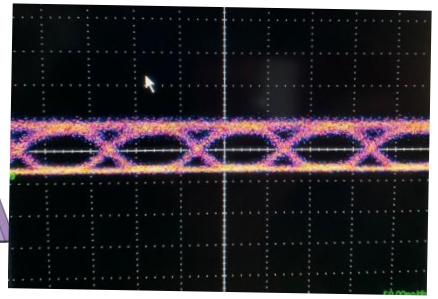
Aug. 30, 2021 — Analog photonic solutions offer unique opportunities to address complex computational tasks with unprecedented performance in terms of energy dissipation and speeds, overcoming current limitations of modern computing architectures based on electron flows and digital approaches.







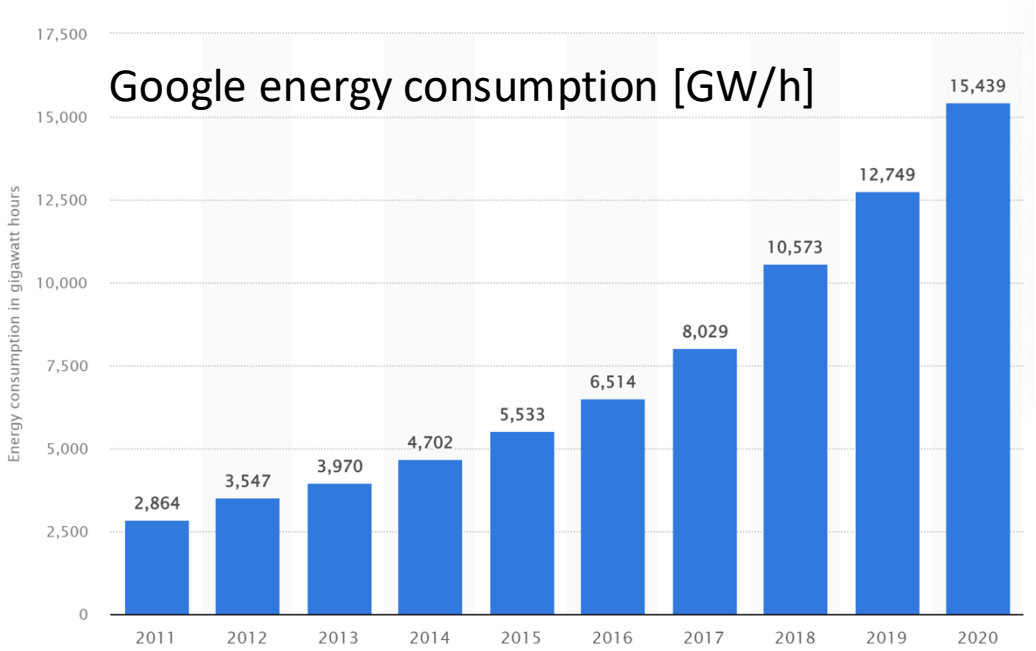




sapere che ogni azione digitale ha una conseguenza sull'ambiente. Un esempio? Dispositivi, server e reti producono il doppio di CO_2 del traffico aereo"

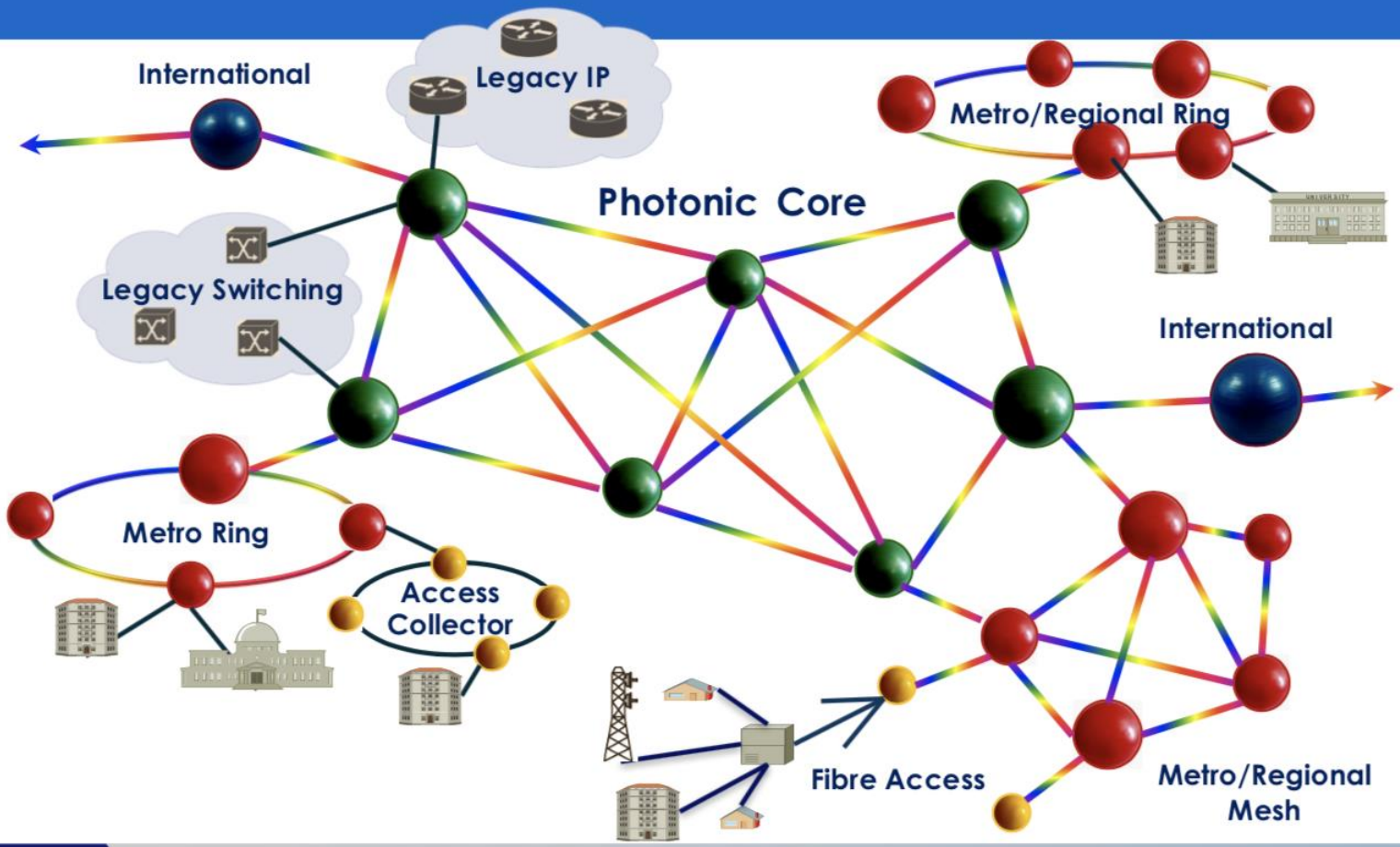
9th February 2021

Roberto Cingolani, nuovo ministro della transizione ecologica



and 6G even worst … 10^7 internet connection per km^2 !

Photonics is the network enabler



Photonic Core

Wavelength-Division Multiplexing (WDM) everywhere; optical switching at nodes (avoid optical-electrical conversion); tunability; increased spectral efficiency; increased bit rate per carrier: 100, 200, 400 Gb/s… All drivers for photonic integration

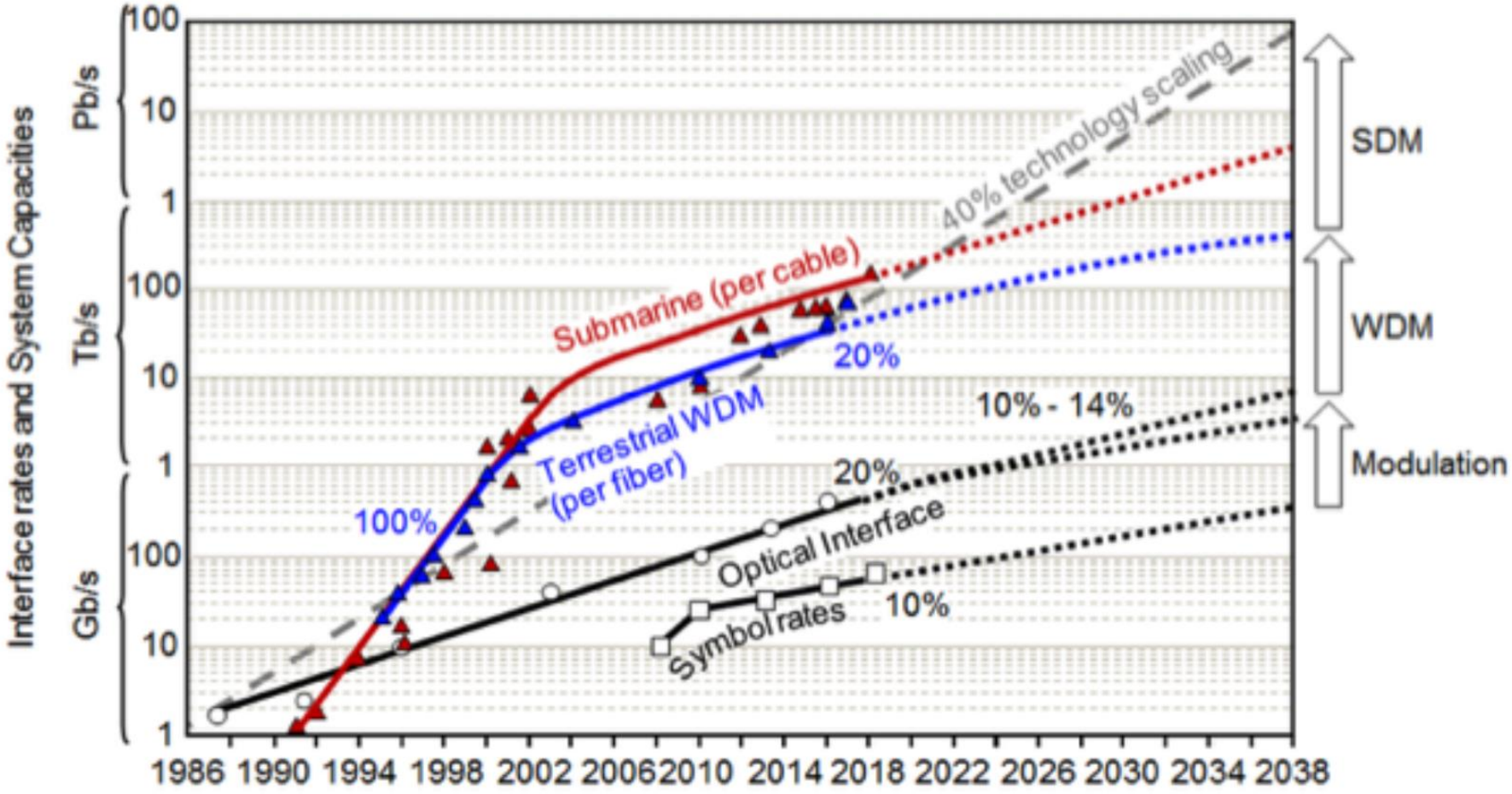


Fig. 1. Evolution of commercial optical transmission systems over the past 30 years and extrapolations for the coming 20 years (after [1]).

MULTIPLEXING….

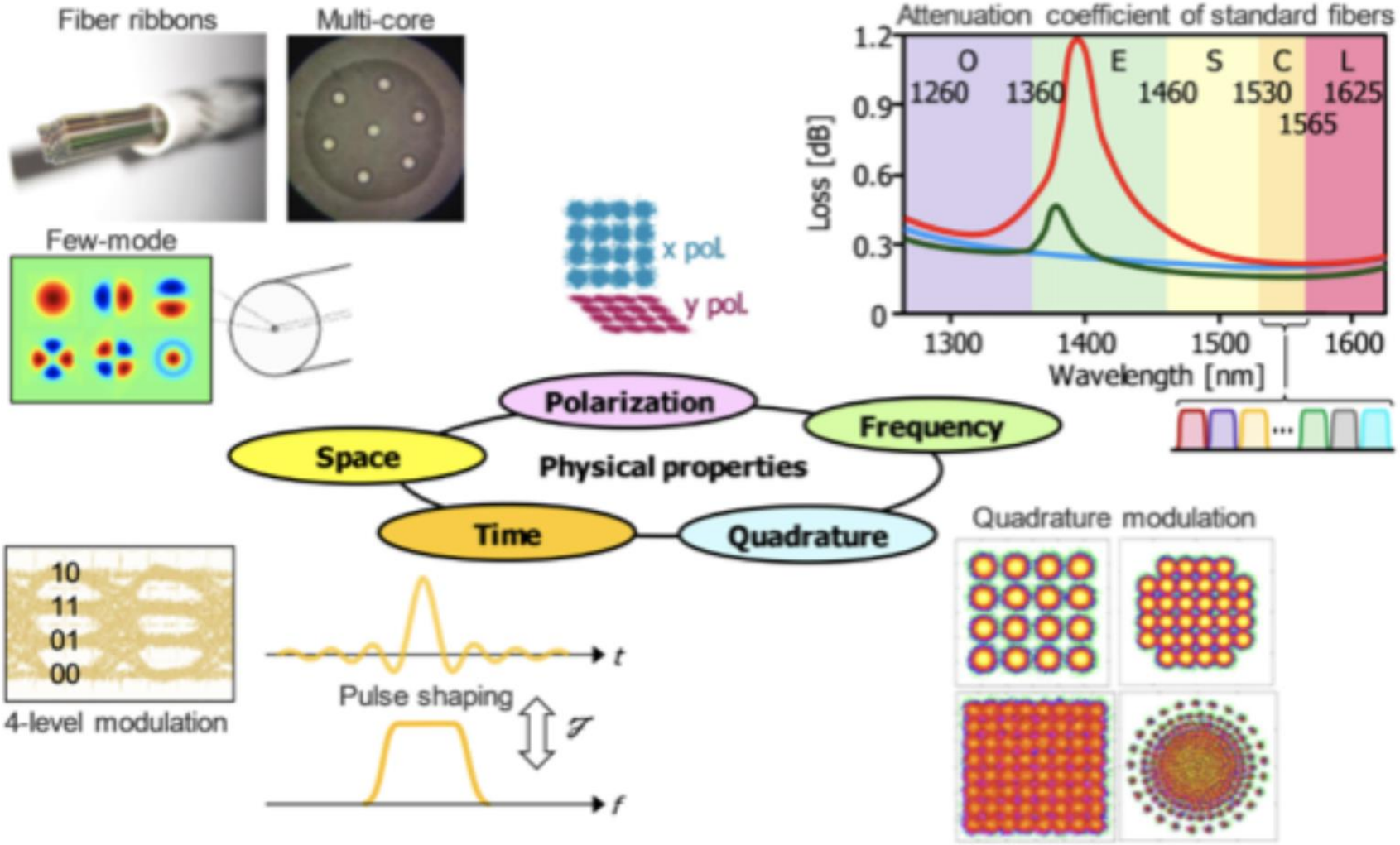
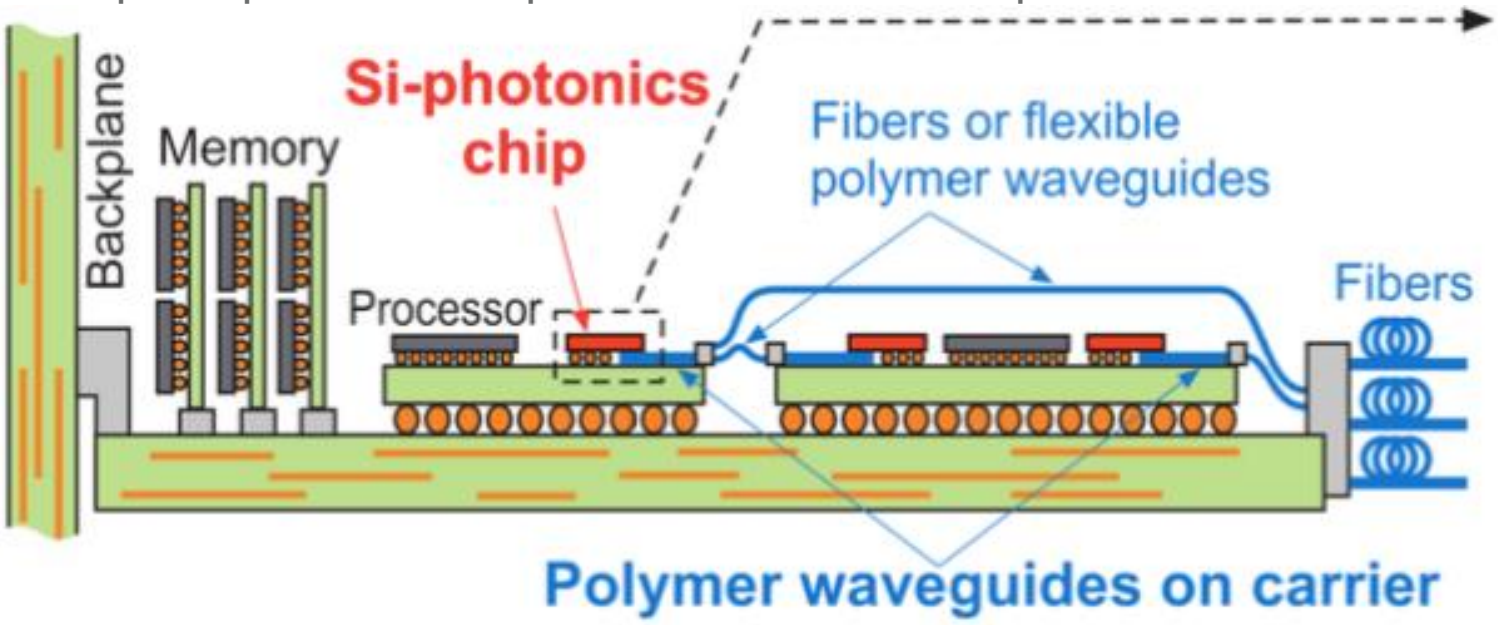


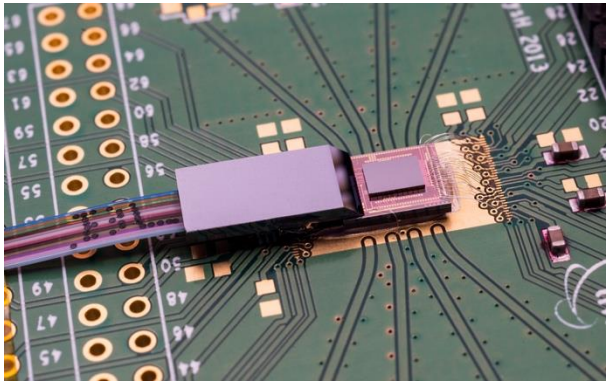
Fig. 15. Physical dimensions available for modulation and multiplexing in optical communications. (Figure after [198].)

Photonics is pervasive ....

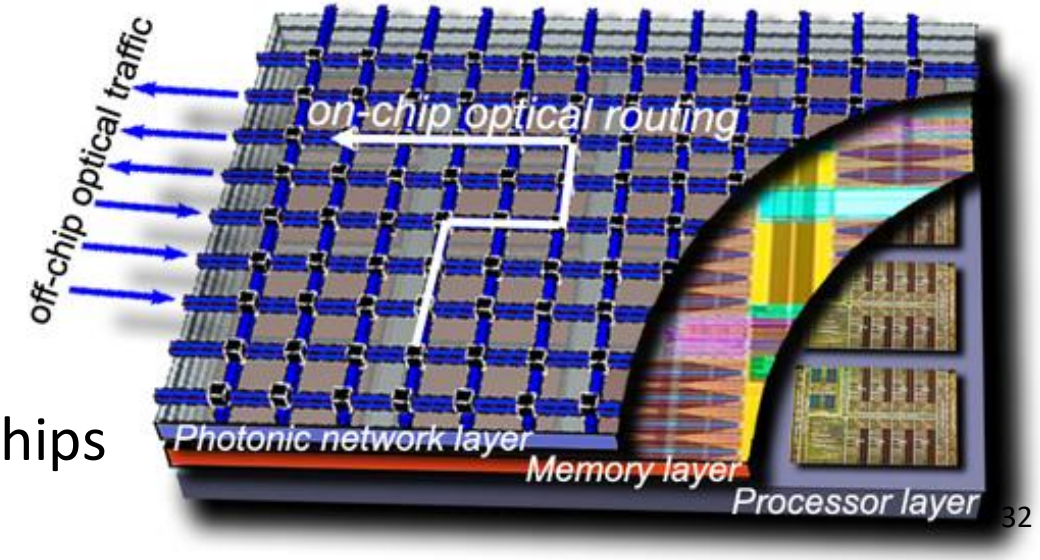
Graphics processor chips have on- and off-chip bandwidth 1.4 Tb/s



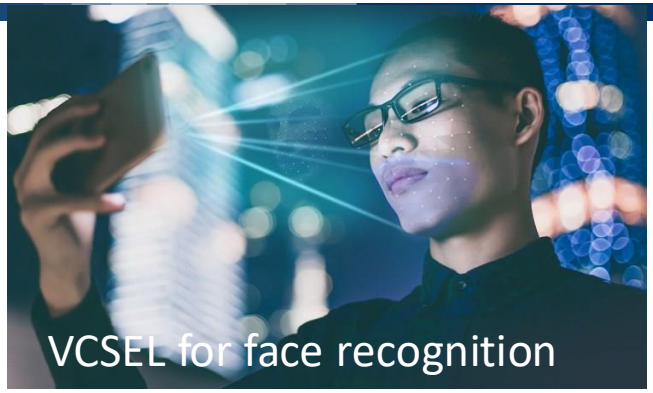
On Boards



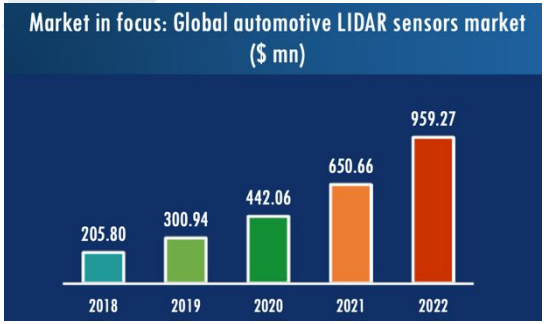
On Chips

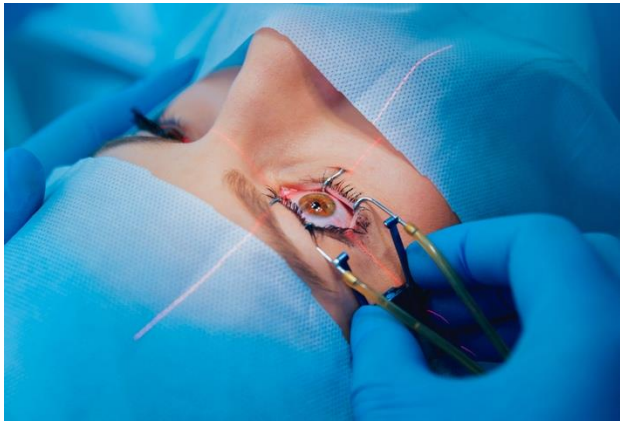


Photonics is pervasive ....



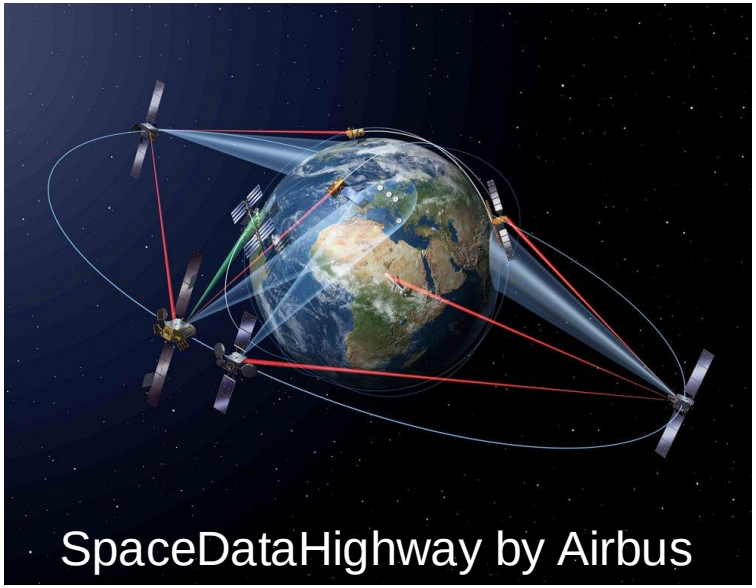




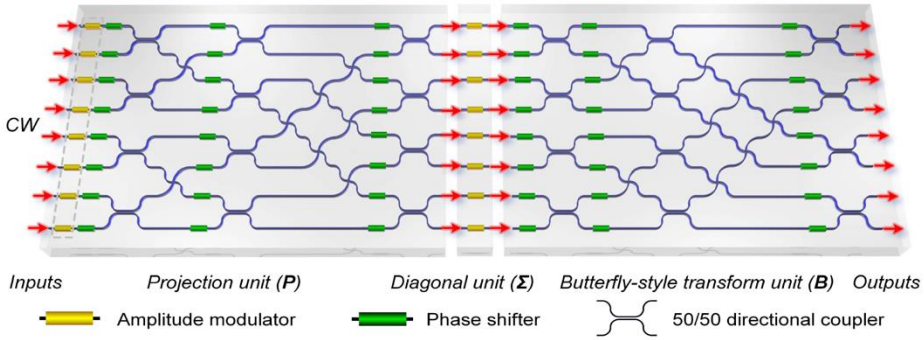


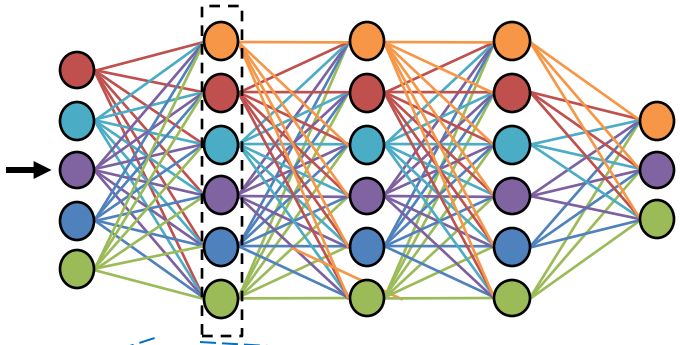
Medical, diagnostic,
optogenetic, sensors …

https://www.epic-assoc.com/about-epic/



Photonic computing, neural networks, AI engines …





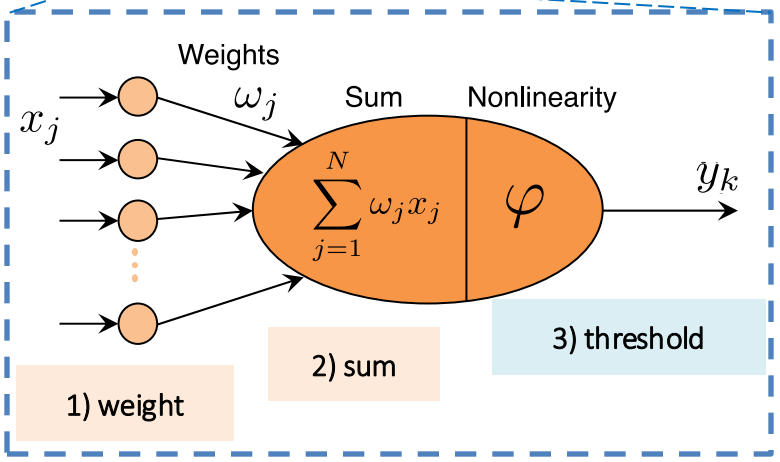


Image recognition

Speech recognition

Singular value decomposition

Matrix Vector Matrix multiplication

Beam forming Networks

MIMO systems

Any algebraic calculation …

Collaboration with Stanford University

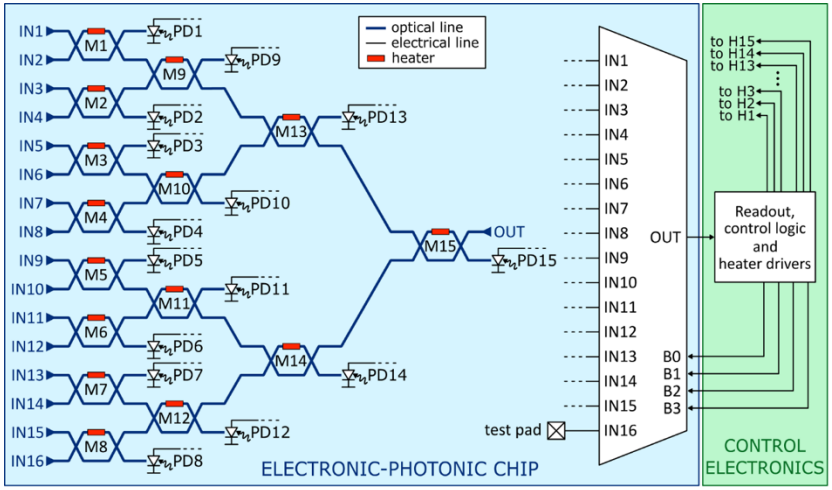
NANOPHOTONICS

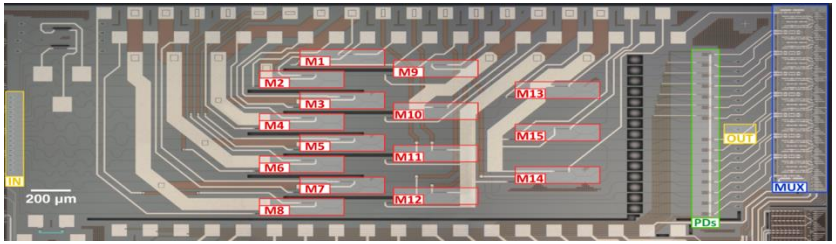
Experimentally realized in situ backpropagation for deep learning in photonic neural networks

Sunil Pai[1*†], Zhanghao Sun[1], Tyler W. Hughes[2‡], Taewon Park[1], Ben Bartlett[2†], Ian A. D. Williamson[1§], Momchil Minkov[1‡], Maziyar Milanizadeh[3], Nathnael Abebe[1#], Francesco Morichetti[3], Andrea Melloni[3], Shanhui Fan[1], Olav Solgaard[1], David A. B. Miller[1]

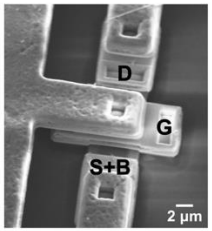
Science, 380, 398–404, 28 Apr 2023

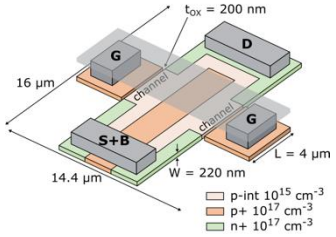
Control electronics in a zero-change silicon photonic platform

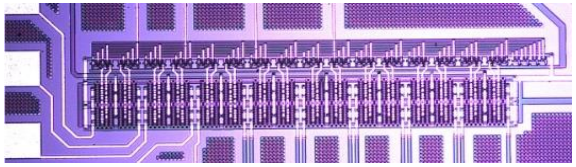




Monolithic electronic de-multiplexer (596 nMOS, 84 pMOS)
Sequential control of actuators and readout of integrated PDs



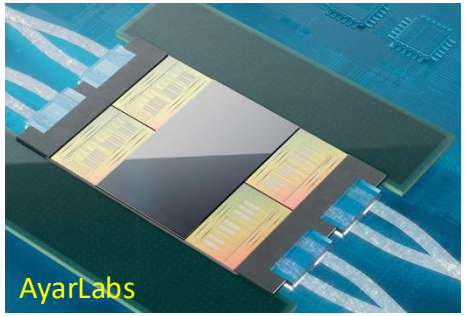


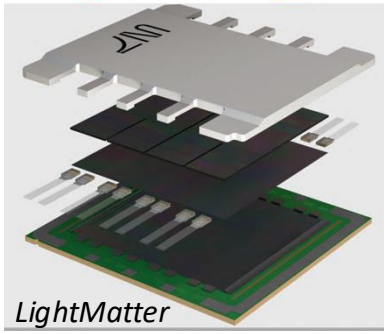


F. Zanetto, *Laser Photonics Rev.* **2023**, 2300124

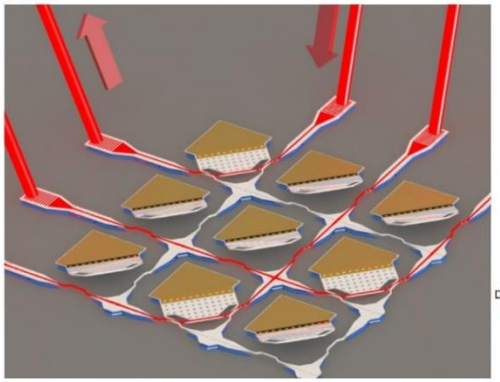
Prof. M. Sampietro
DEIB

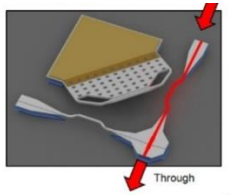
Co-Packaged Optics

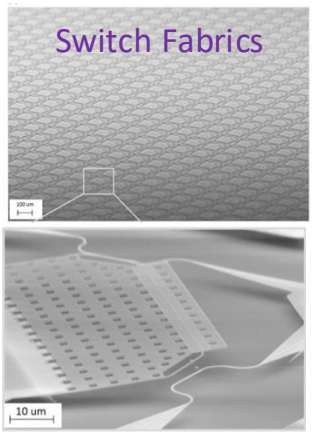




MEMS & Photonics







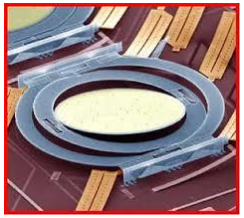
Ming C. Wu, University of California, Berkeley, USA
Nature **volume 603**, pages 253–258 (2022)

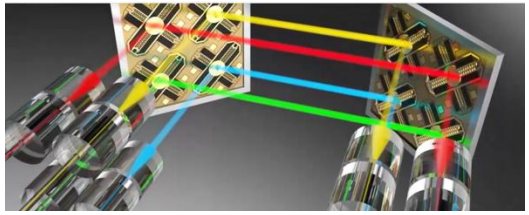
Datacenters vs FTTH





384x384

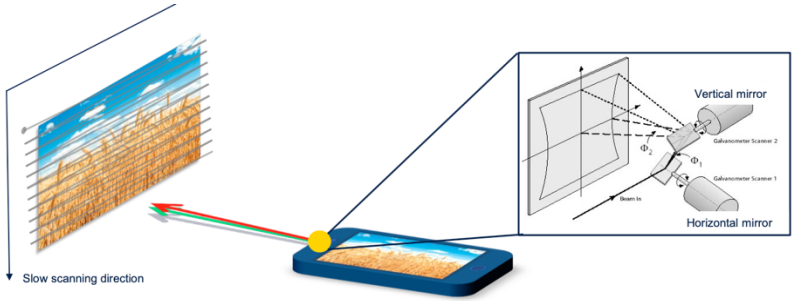


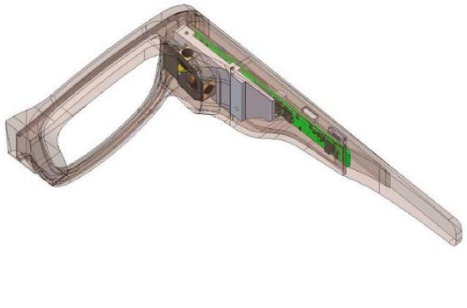


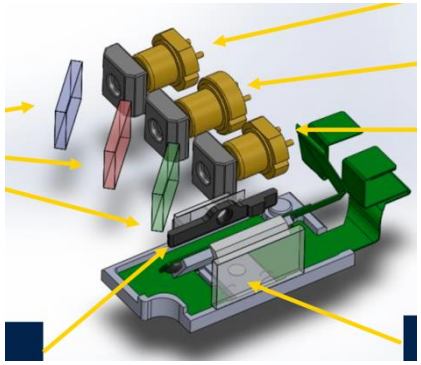
FTTH Council EU

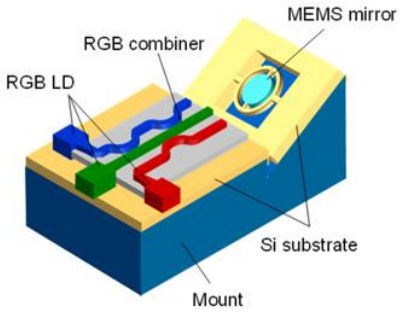
- FTTH/FTTB EU coverage: 70 %
- *Italy forth in Europe*
- Very High costs of cabinets management
- MEMS based switch fabrics ??

Laser Beam Scanning - RGB

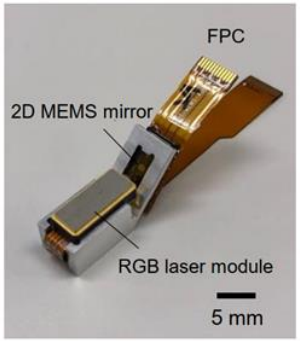


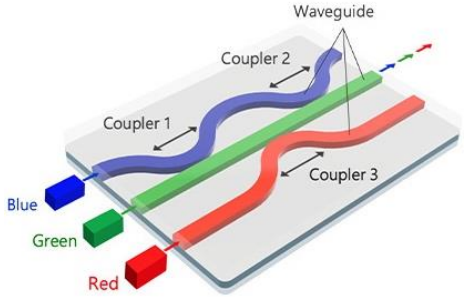




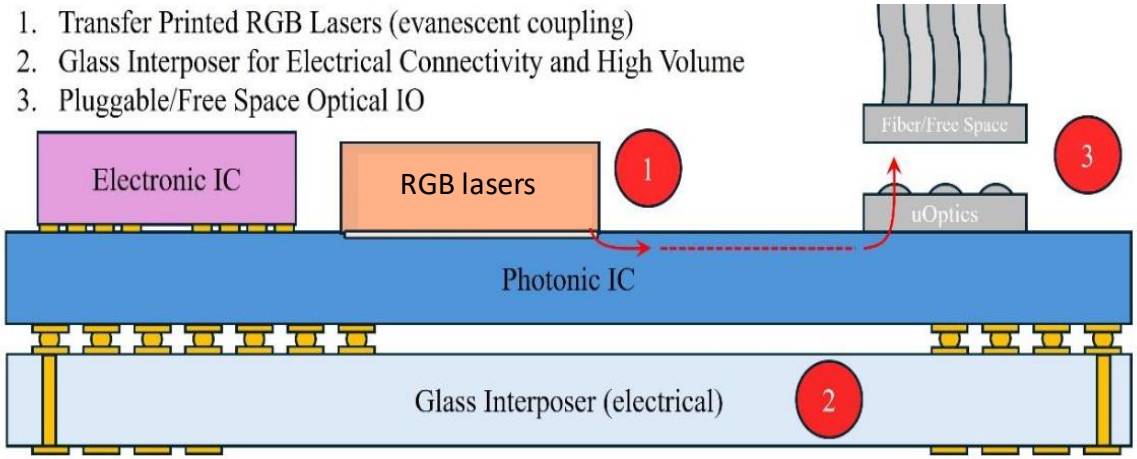


University of Fukui





1. Transfer Printed RGB Lasers (evanescent coupling)
2. Glass Interposer for Electrical Connectivity and High Volume
3. Pluggable/Free Space Optical IO



TOWARDS 2020 – PHOTONICS DRIVING ECONOMIC GROWTH IN EUROPE

Multiannual Strategic Roadmap 2014–2020



PHOTONICS21

https://www.photonics21.org/

Europe's age of light!

How photonics will power growth and innovation

2020-2030

Download from:

https://www.photonics21.org

PHOTONICS21

PHOTONICS 21

Photonics

A Key Enabling Technology for Europe

The European Technology Platform Photonics21 represents the photonics community of industry and research organisations. Jointly with the European Commission our members develop and implement a common photonics strategy in a Horizon2020 **Public Private Partnership (PPP)** to spur growth and jobs in Europe.

Photonics21 – Photonics PPP

Annual Activity Report 2017

PHOTONICS PUBLIC PRIVATE PARTNERSHIP

POLITECNICO DI MILANO

Europe's age of light!

How photonics will power growth and innovation

2020-2030

Download from:

https://www.photonics21.org

PHOTONICS 21

2. Photonics Research and Innovation Challenges 18

2.1 Information & Communication 18

2.2 Industrial Manufacturing & Quality 35

2.3 Life Science & Health 41

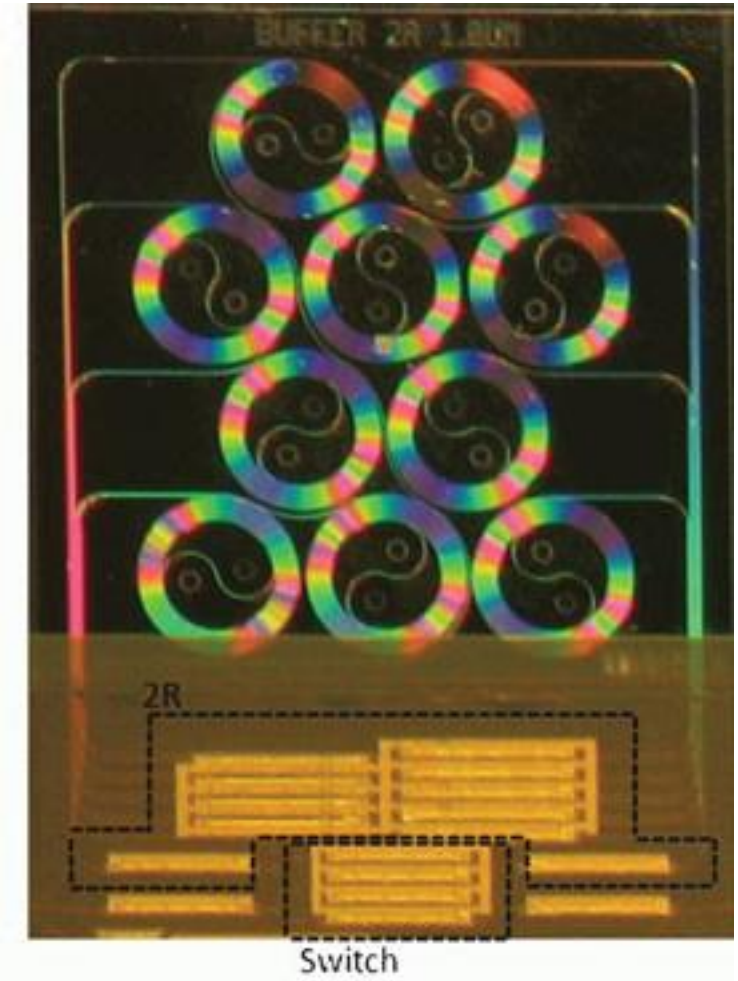
2.4 Emerging Lighting, Electronics & Displays 49

2.5 Security, Metrology & Sensors 60

2.6 Design and Manufacturing of Components & Systems 70

2.7 Education, Training & Disruptive Research 83

2.1 Information & Communication







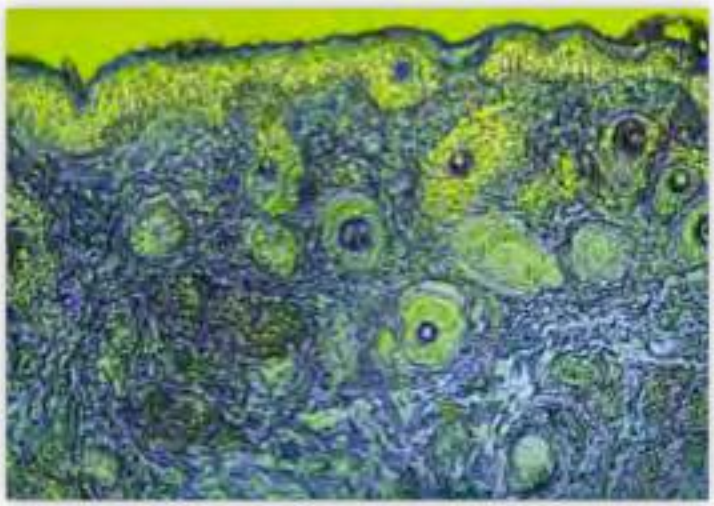
2.2 Industrial Manufacturing & Quality



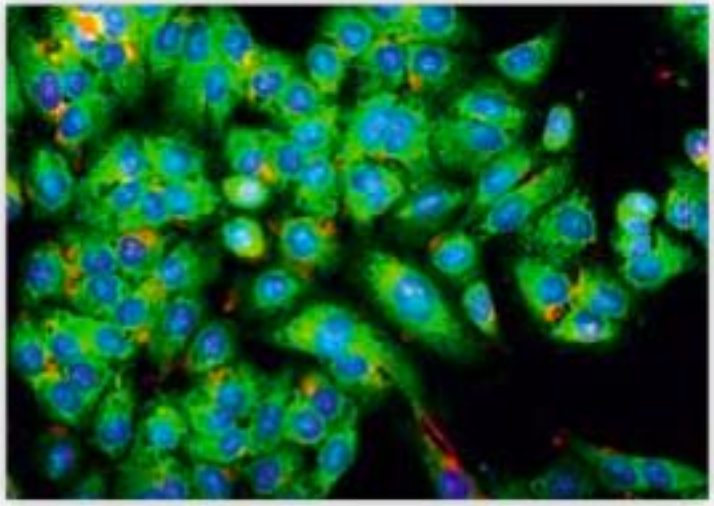


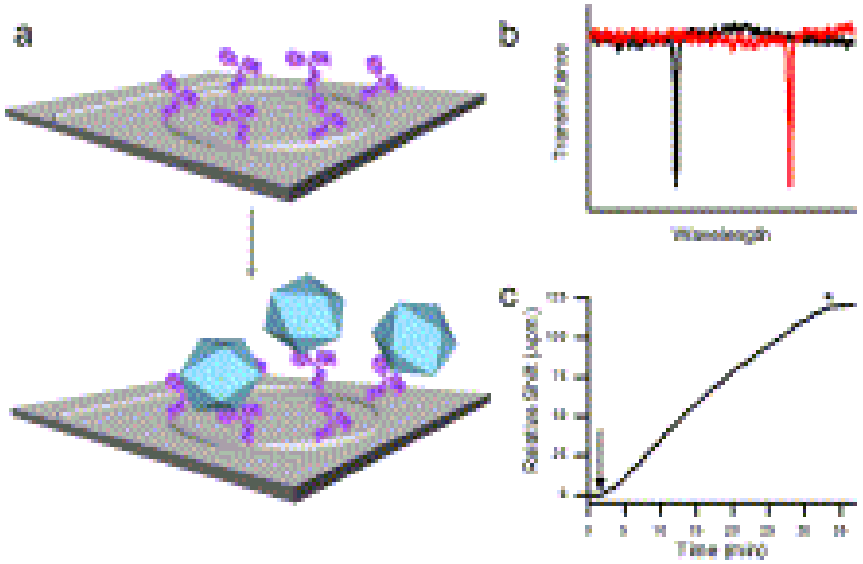


2.3 Life Science & Health









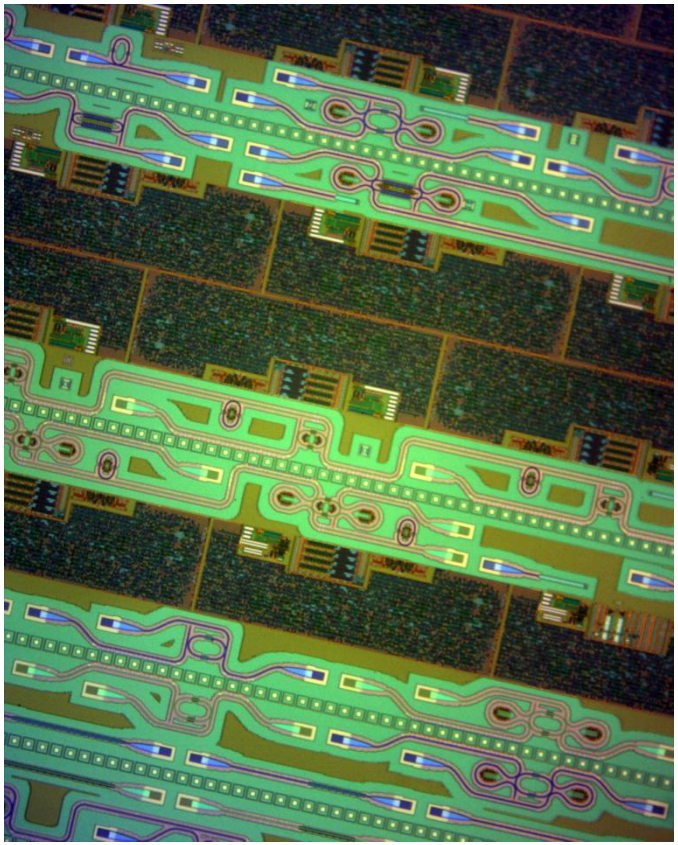
2.4 Emerging Lighting, Electronics & Displays



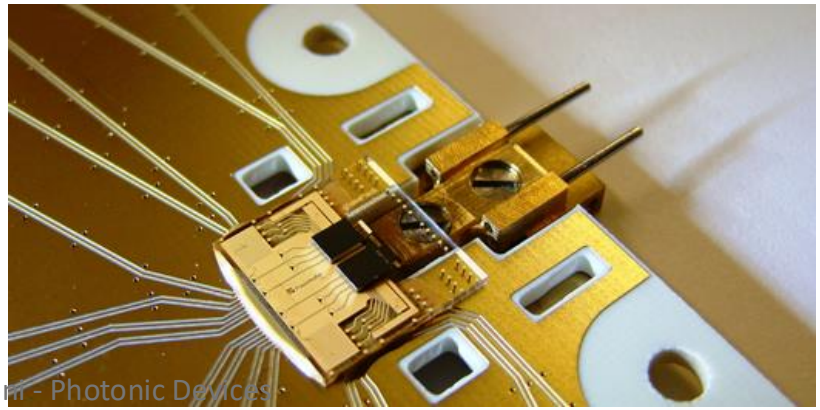




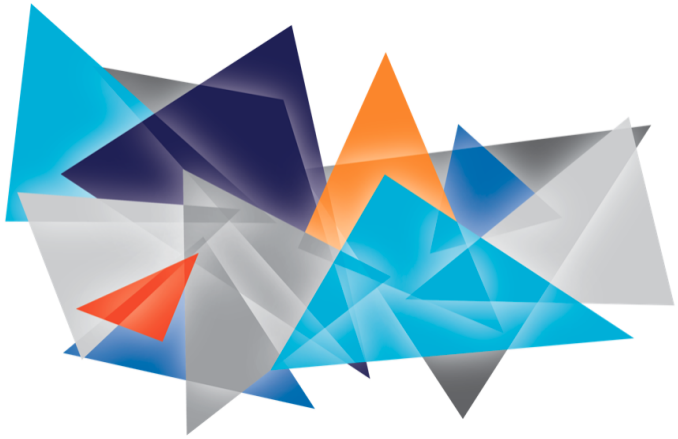
2.6 Design and Manufacturing of Optical Components and Systems











Photonics

A Key Enabling Technology for Europe

The European Technology Platform Photonics21 represents the photonics community of industry and research organisations. Jointly with the European Commission our members develop and implement a common photonics strategy in a Horizon2020 **Public Private Partnership (PPP)** to spur growth and jobs in Europe.

Work Groups

Work Group 1
Digital Infrastructure

Work Group 2
Manufacturing

Work Group 3
Health

Work Group 4
Climate, Mobility & Energy

Work Group 5
Safety, Security, Space & Defense

Work Group 6
Agriculture & Food

Work Group 7
Core Photonics

POLITECNICO MILANO 1863

PHOTONICS21

EPIC European Photonics Industry Consortium

Europe is the world's 2nd largest producer of photonics products

European photonics at a glance

Investment

Nearly **€10Bn** – In 2015 European Photonics Industry pledged €9.6Bn to Innovation (R&D spending and Capex)

R&D intensity in the Photonics industry amounts to nearly **10%**

Photonics industry is leveraging public investment in the PPP projects by a factor of **4.3**

Industry participation in Horizon 2020 PP projects increased to **45%** compared to 35% in FP7

7 Prototyping and Pilot Manufacturing services to help end user industry to speed up product development

€49M invested by the EC in the PPP Pilot manufacturing services (Pilot lines)

67 projects started so far under the frame of Horizon 2020 with a public investment of a total of **€278M** EC funding over the first 3 years of Horizon 2020: 2014-2016

Market

The European Photonics market amounts to **€69Bn** per annum

European Photonics Production has increased by over **62%** over the last 10 years

The European Photonics Production has grown with an average CAGR of **5%** since 2005

European Photonics Production Growth rate is more than **3.5** higher than EU GDP Growth rate

50% global market share for European Photonics for Production Technology

35% global market share for European Photonics for Optical Measurement & Image Processing

32% global market share for European Photonics for Optical Components and Systems

Global Photonics Industry grew from **€228Bn** in 2005 to **€447Bn** in 2015

Global Photonics market expected to amount to **€615Bn** in 2020

The photonics sector is **growing twice as fast** as the global GDP

People

19,000 new jobs created in the European Photonics Industry

Forecast: **42,000** new jobs could be created by 2020

9 PPP Projects explicitly devoted to promoting Research, Education & Training in Photonics

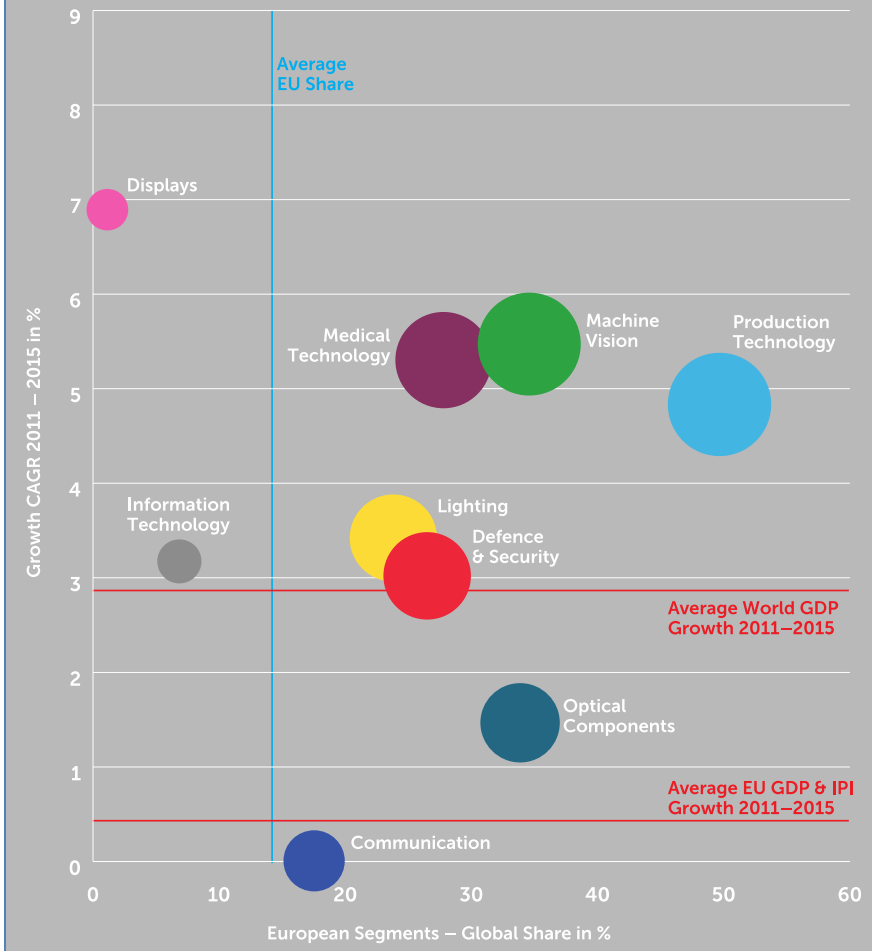
More than **1700** Photonics companies and research organisations back the Photonics PPP.

More than **3000** registered personal members in the platform Photonics21

About **1000** attendees in Photonics PPP strategy development workshops and meetings

Around **300,000 people** in Europe are employed directly in the photonics sector

Global leadership position in European core photonics segments

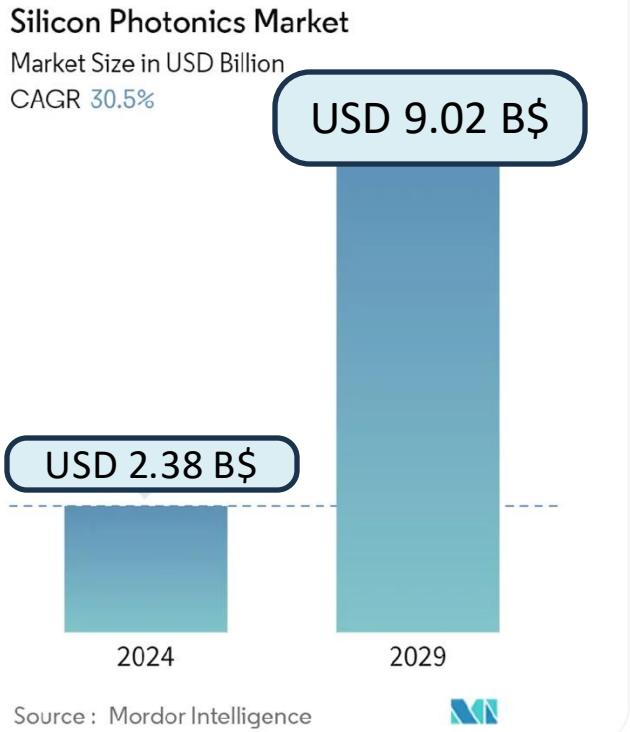


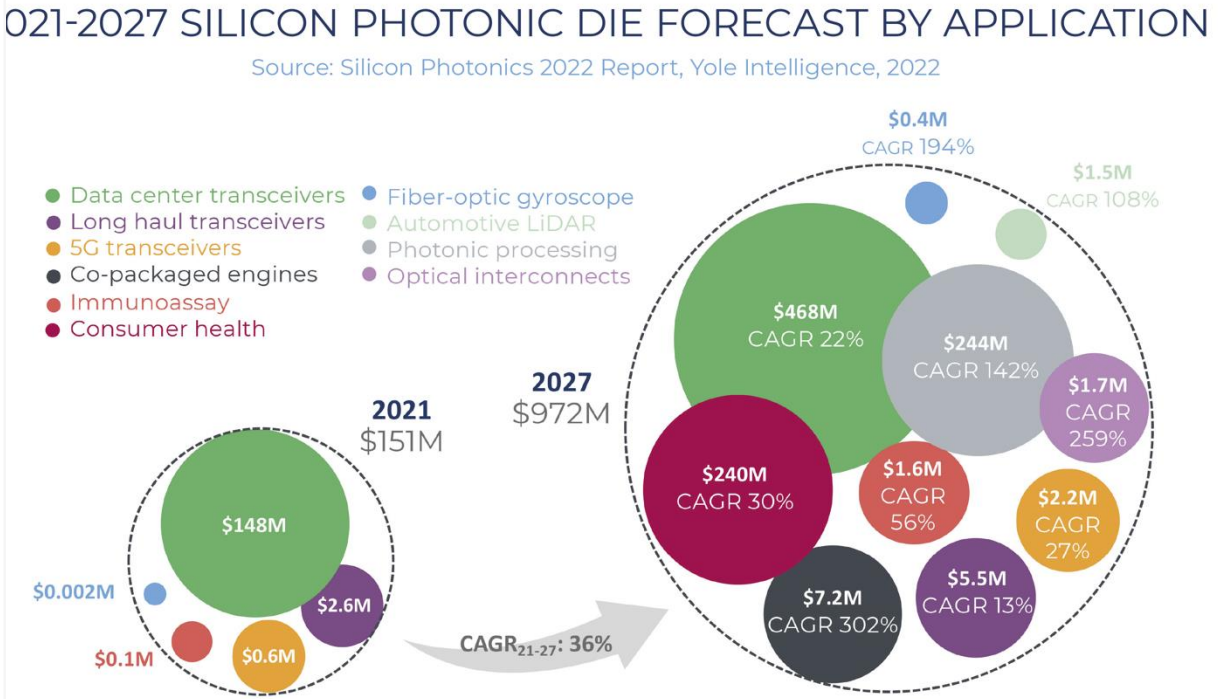
Source: Optech Consulting Market Research Study 24.1.2017

Silicon Photonics: roadmap and markets

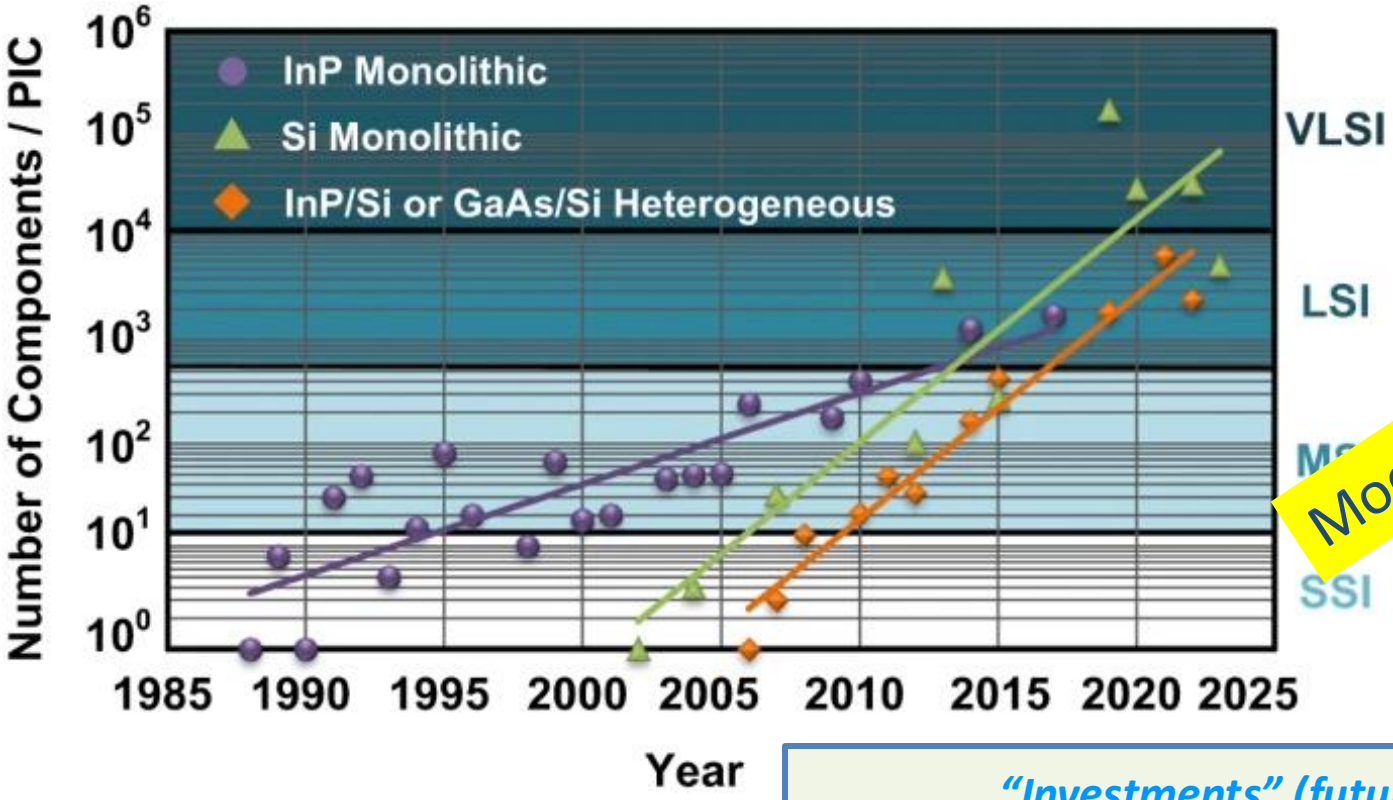








Roadmapping



Moore law in photonics??

Shekhar, S. *et al.* Roadmapping the next generation of silicon photonics. *Nature Commun* **15**, 751 (2024)

"Investments" (future directions)

- CONTROL: Electronic/Photonic integration
- Low power Actuators
- Wafer level - Volume testing & Calibration
- Multi-physics holistic codesign
- Co-Packaged Optics (Heterogeneous)
- General purpose devices (FPGA)

To mimic the success of electronic….

Analog → Digital

No, Photonics is analog by nature …..

Scaling

Waveguide size ≃ λ/refractive index…

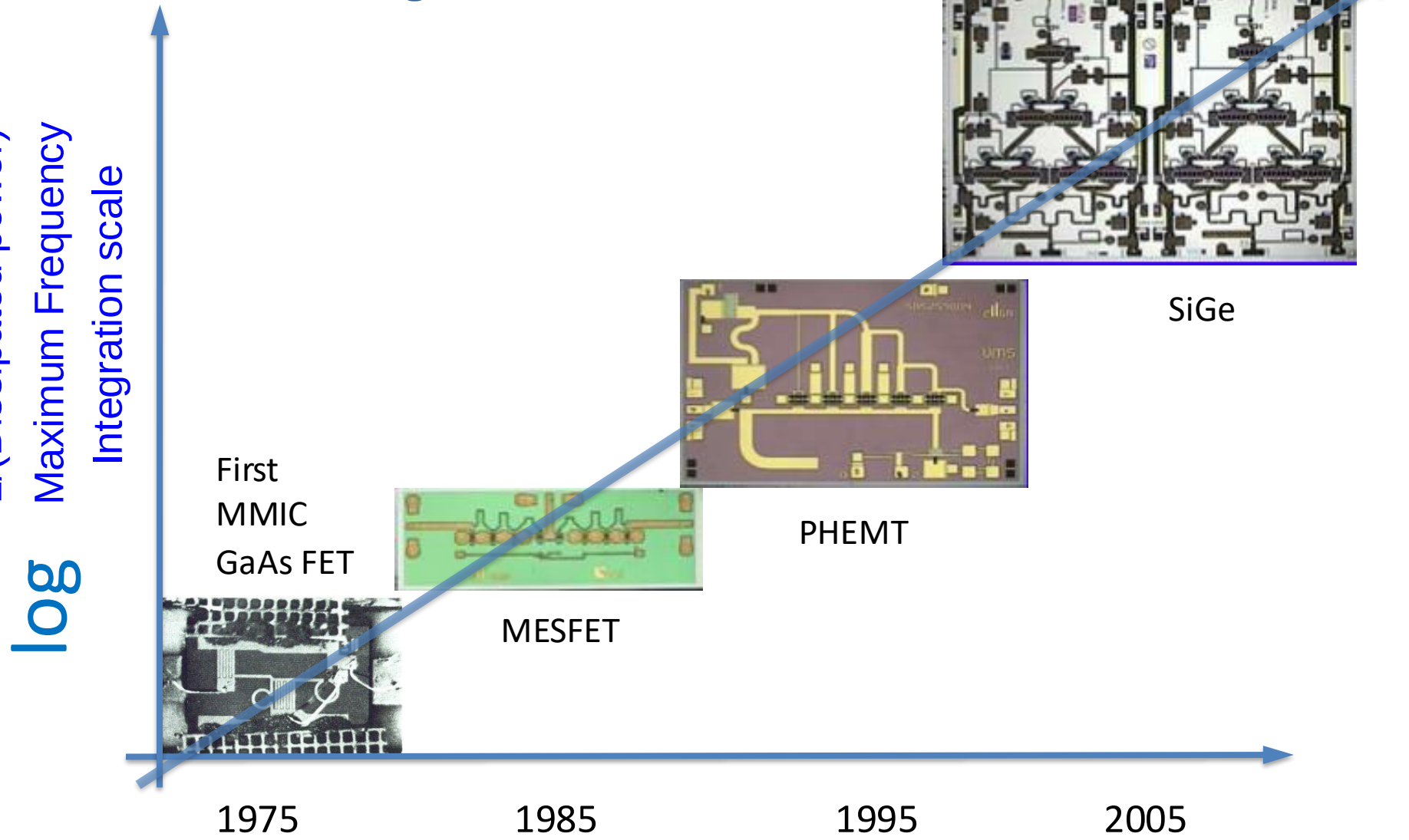
(no scaling for photonics)

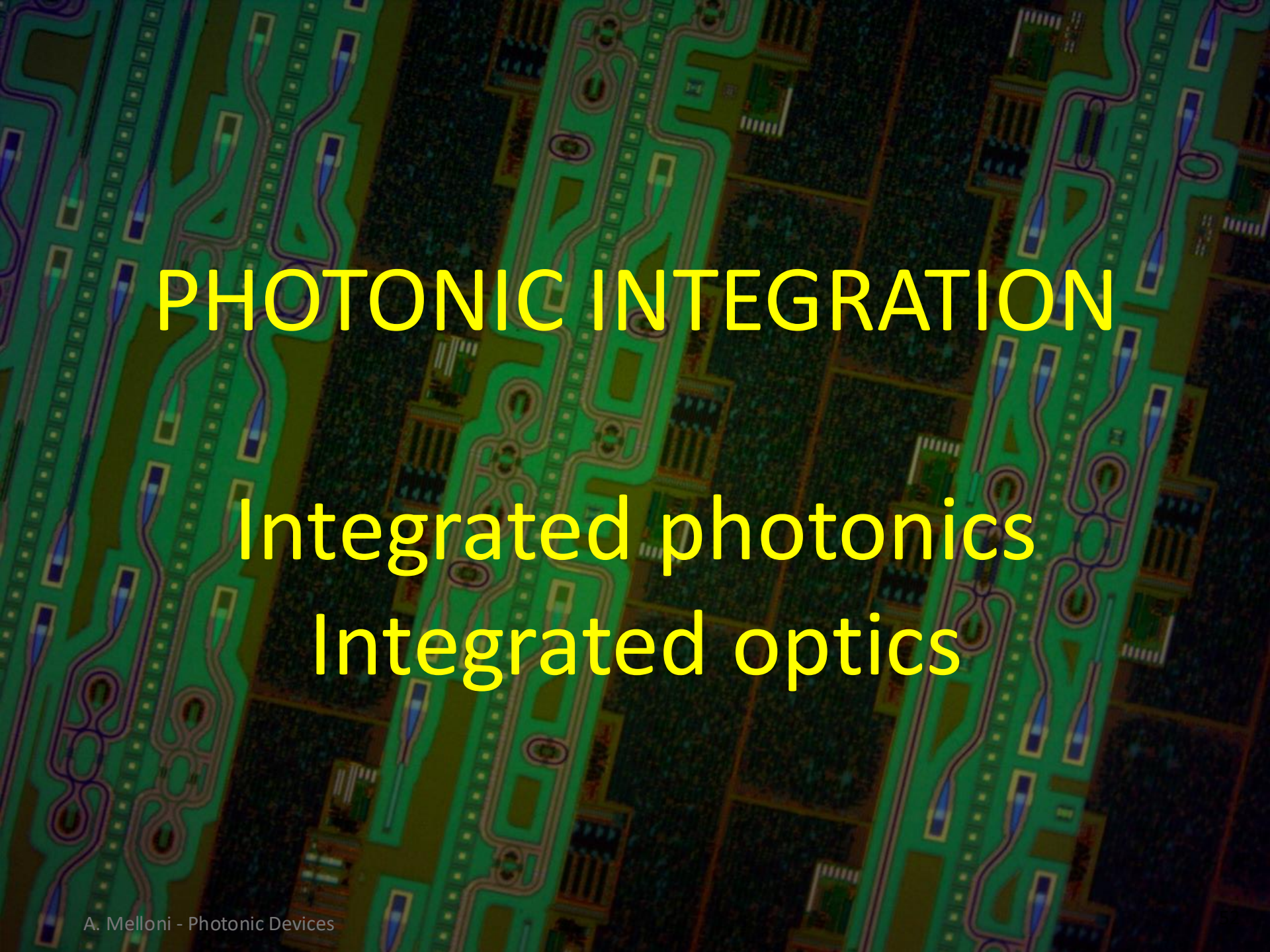
Generic foundries

coming now !

Photonics is analog …

Microwave is analog …




PHOTONIC INTEGRATION
Integrated photonics
Integrated optics

From research to commercial technology

THE BELL SYSTEM
TECHNICAL JOURNAL

DEVOTED TO THE SCIENTIFIC AND ENGINEERING
ASPECTS OF ELECTRICAL COMMUNICATION

Volume 48 September 1969 Number 7

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Integrated Optics: An Introduction

By STEWART E. MILLER

(Manuscript received January 29, 1969)

Dielectric Rectangular Waveguide and Directional Coupler for Integrated Optics

By E. A. J. MARCATILI

(Manuscript received March 3, 1969)

Optical Waveguide of Macroscopic Dimensions in Single-Mode Operation*

ROBERT A. KAPLAN
Wheeler Laboratories, Inc.
Great Neck, N. Y.

1969

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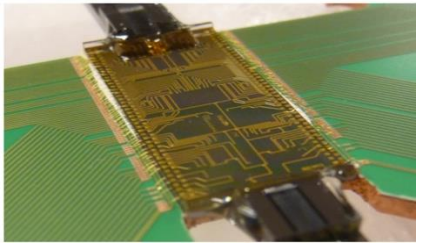
Dielectric Rectangular Waveguide and Directional Coupler for Integrated Optics

By E. A. J. MARCATILI

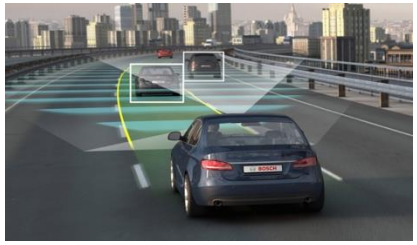
(Manuscript received March 3, 1969)

Optical Waveguide of Macroscopic Dimensions in Single-Mode Operation*

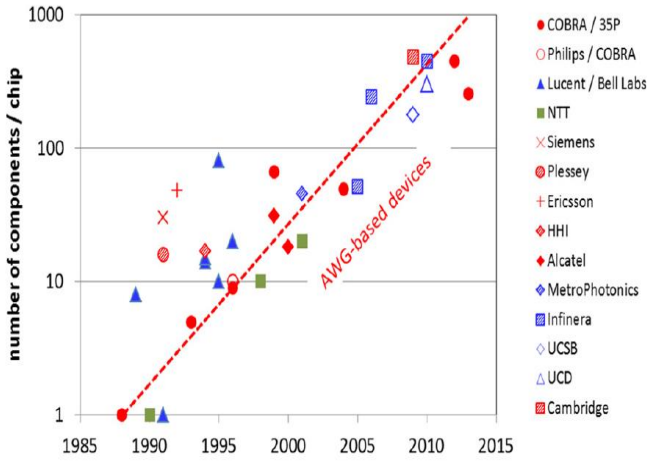
ROBERT A. KAPLAN
Wheeler Laboratories, Inc.
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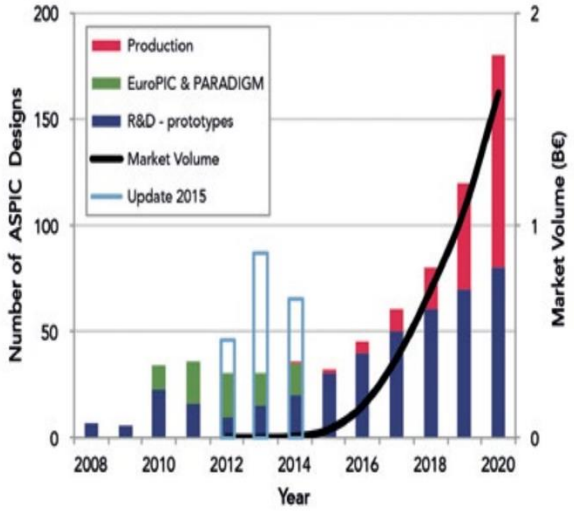












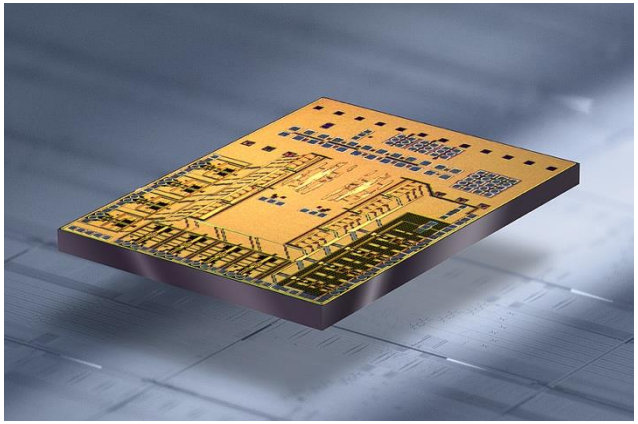
Meint Smit et al, Semiconductor Science and Technology, 2014

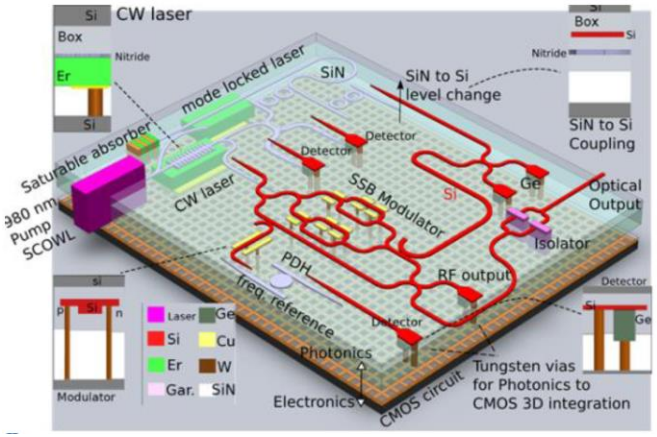
1969

2020

Integrated photonics



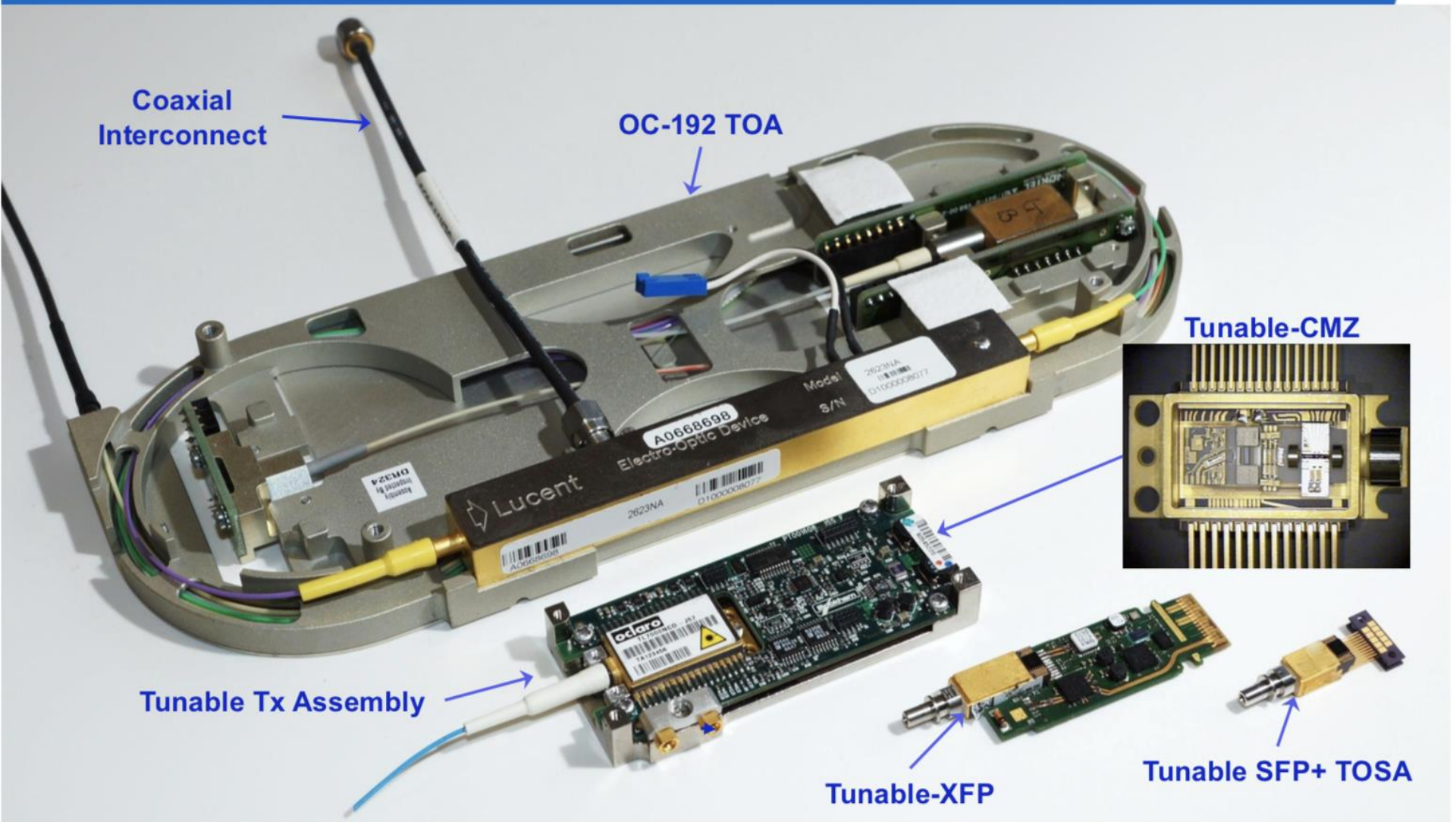






Integrated photonic circuits - PIC

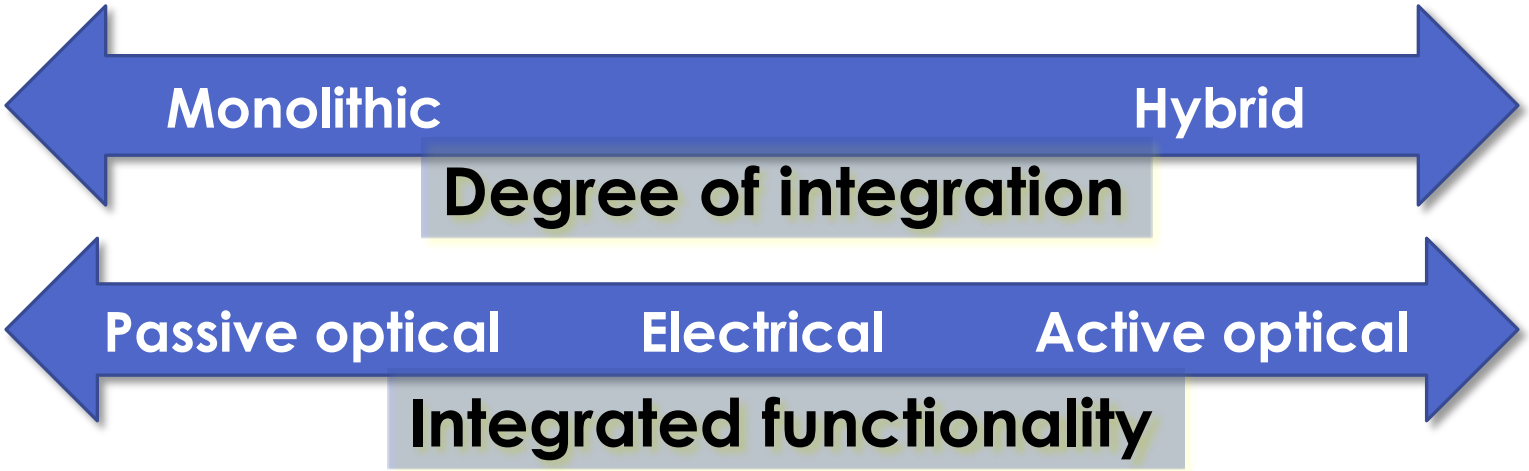
Integrated Photonics



T-SFP+ TOSA has full TOA functionality plus wavelength tunability & control

What do we mean by Integrated Photonics?

- Integrated = put together
- Photonics = optical/electrical components *(sources, modulators, VOAs, mux/demux, splitters/ combiners, detectors, …)*
- Integrated Photonics = putting together a selection of electro-optic components into a single package.



Photonic Integration: Motivation

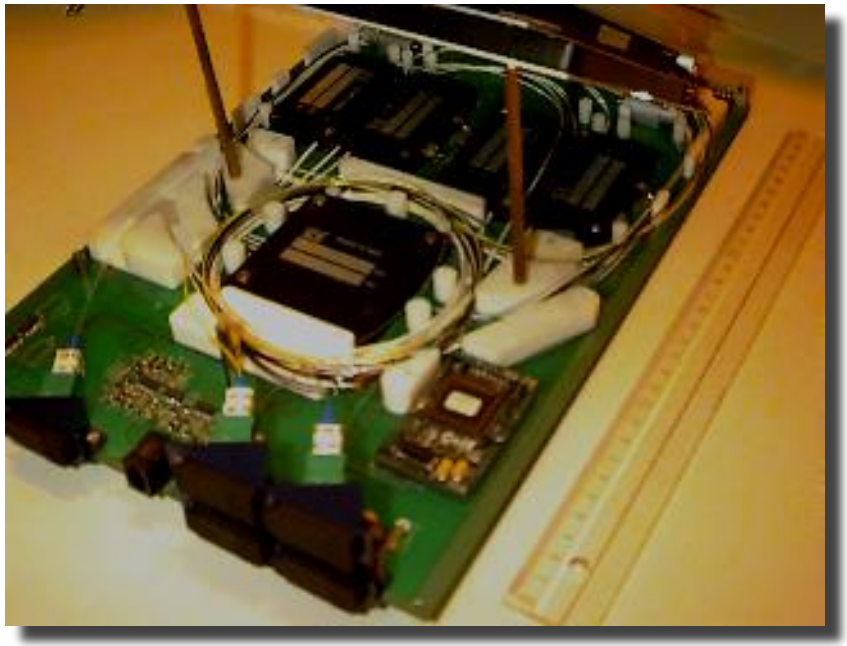
POLITECNICO DI MILANO

- Greatly reduced component cost
  - Monolithic interconnection of device elements
  - Simpler packaging and assembly, standard processes
- High reliability
  - Less interfaces
- High functionality
  - Many more functional elements per chip, higher creativity in design
- High phase stability, excellent device matching
  - Permits interferometric structures
- Robust
  - Single chip designs with minimal optical interfaces are ideal for demanding environments
- Better power efficiency
  - Minimize optical power loss at interfaces between device elements

On the other hand, the capital and R&D investment needed to play at the top level in this field is very large

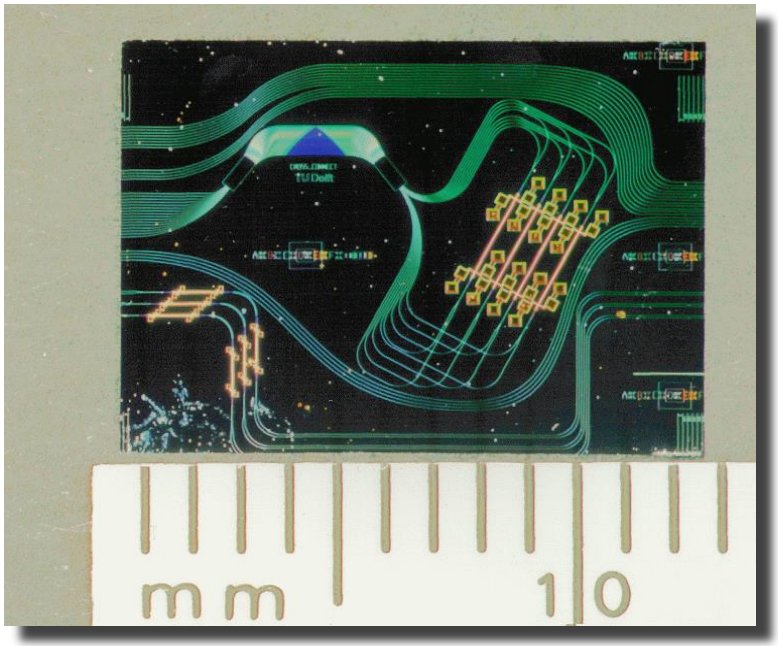
Photonic Integration

reduction of volume and packaging cost



Compact fibre-based cross-connect module 1997

4-channel 2x2 OXC



Photonic Integrated cross-connect chip 1998

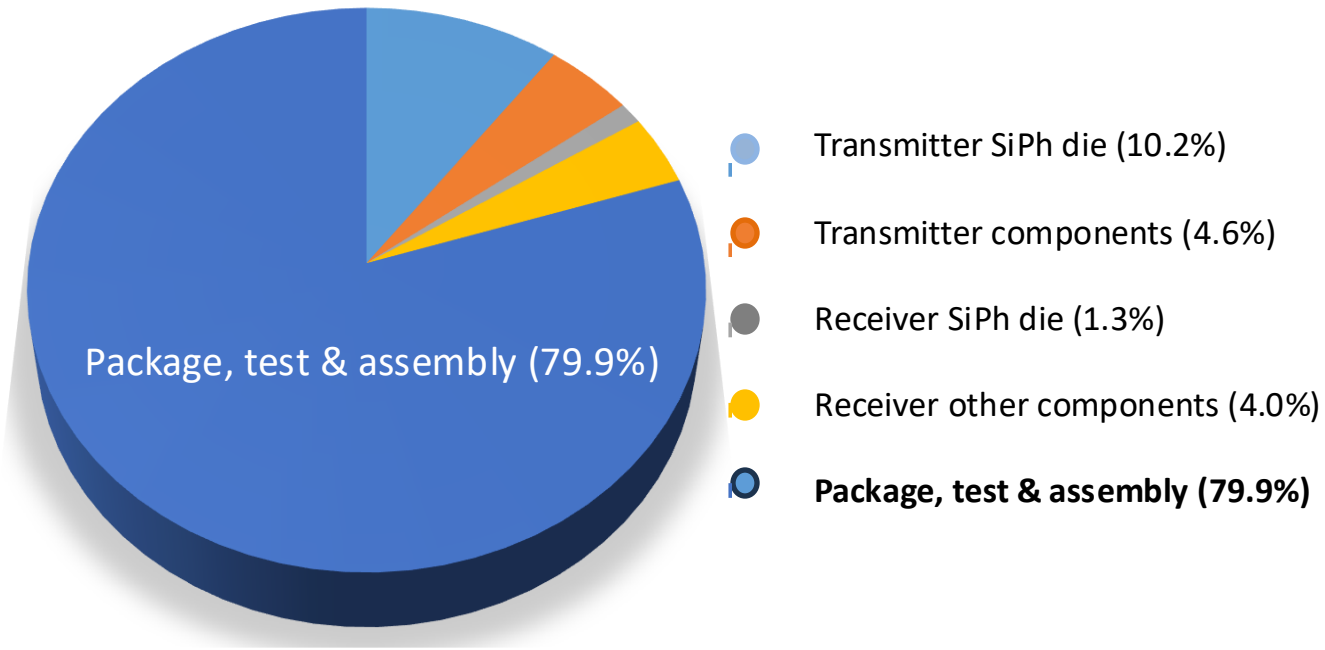
4-channel 2x2 OXC

4-channel 2x2 OXC, 1999

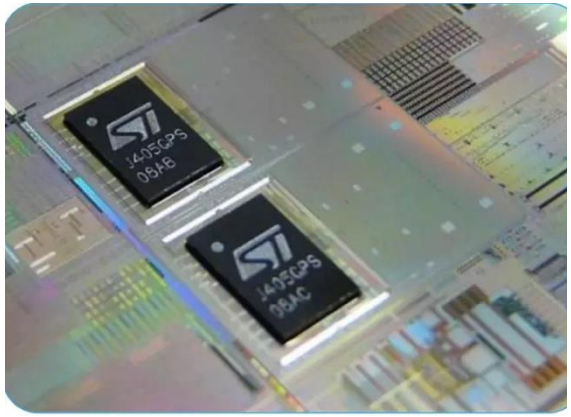


Cost breakdown for SIPh transceiver

SiPh Die: 11.5%



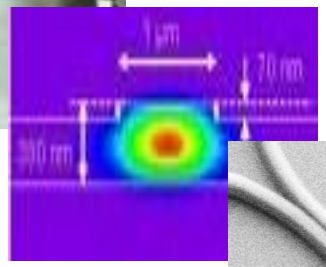
Source:
Silicon photonics economic feasibility and relative cost analysis for automotive Applications, IEEE 802.3cz Multigigabit Optical Automotive Ethernet Interim Task Force

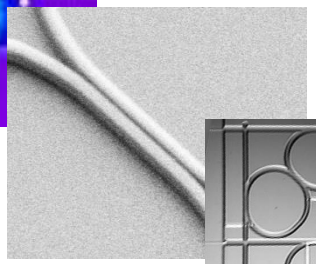


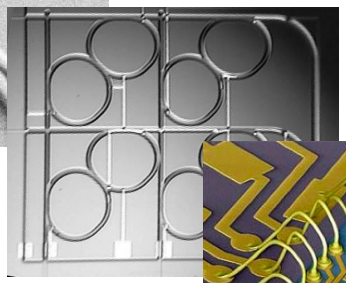
The long march of PICs…

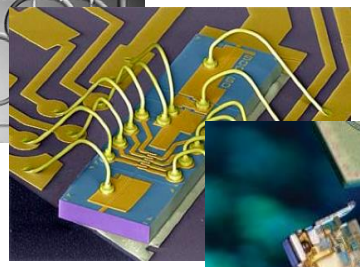


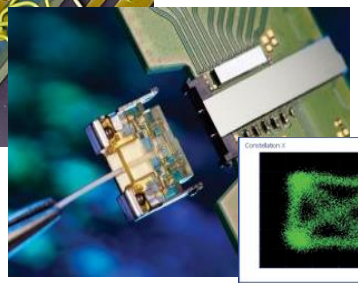


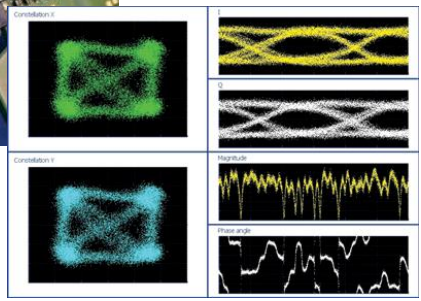












DESIGN

ANALISYS

Generic Foundries

Design houses

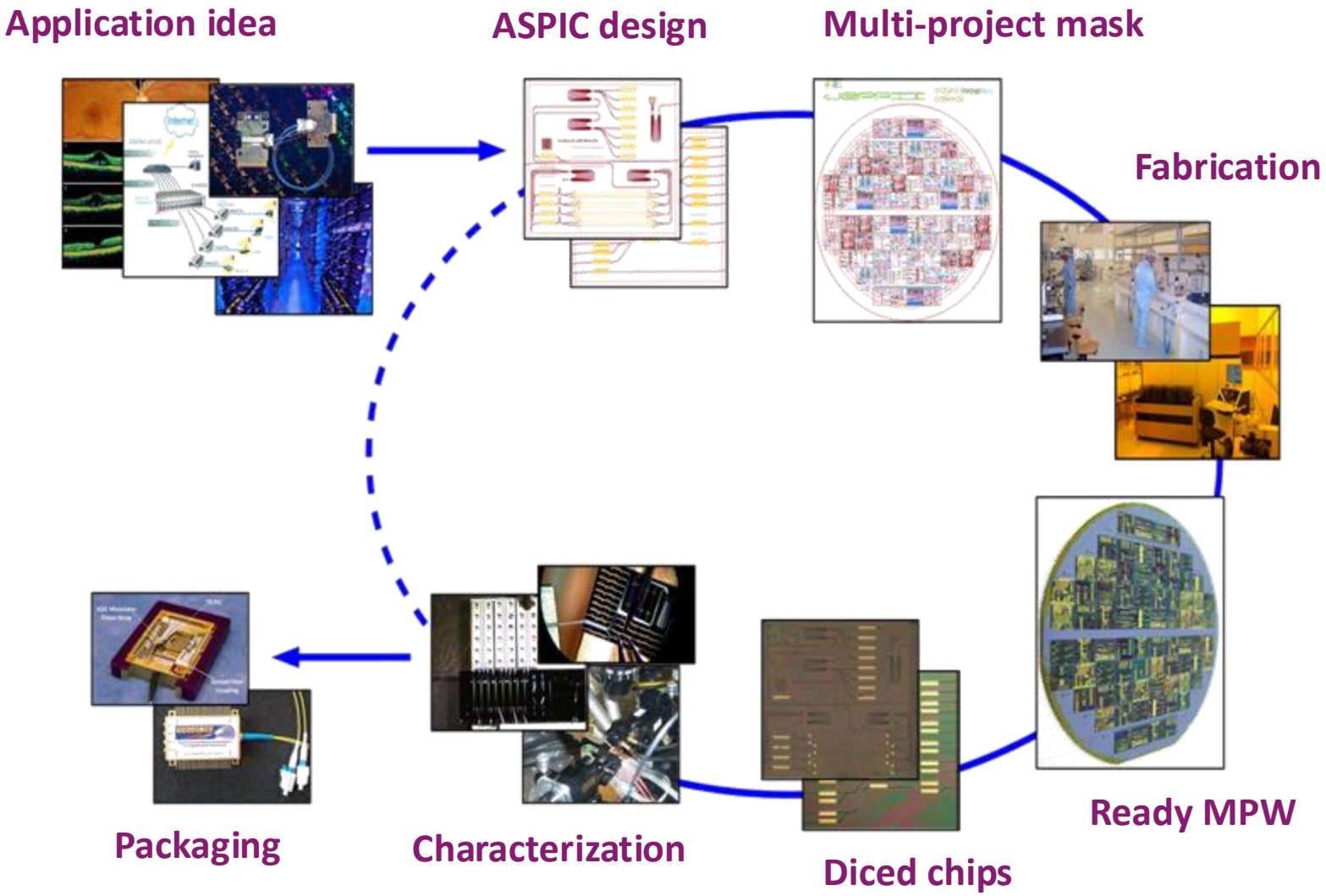
Packaging

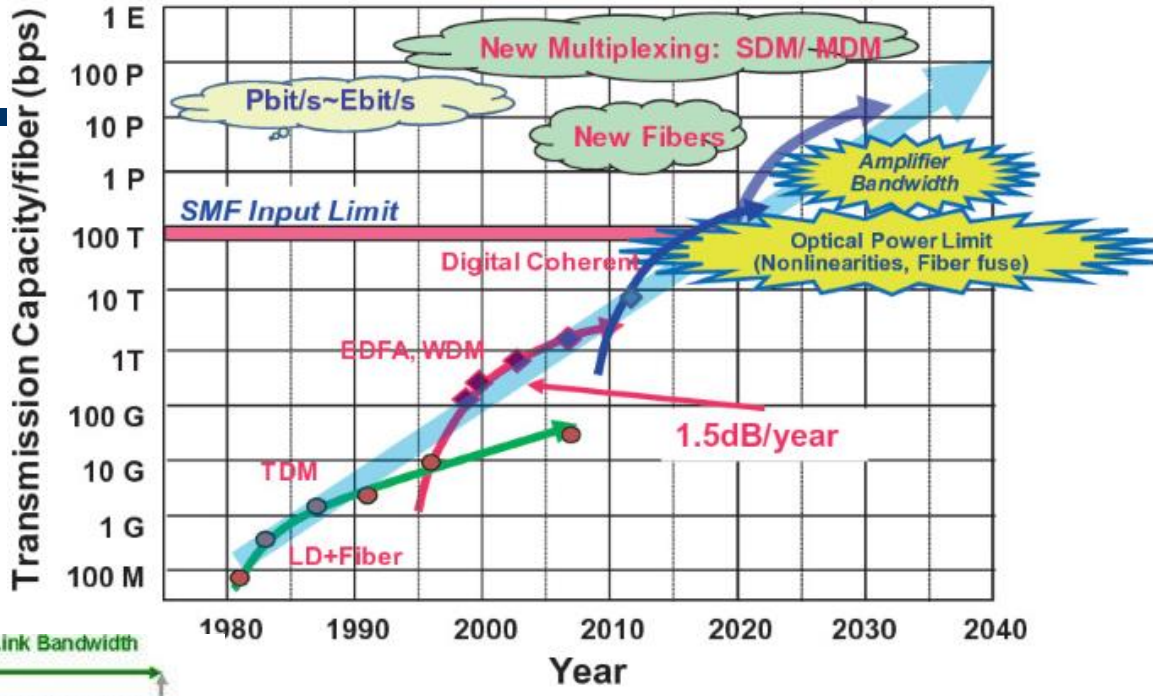
Final Users

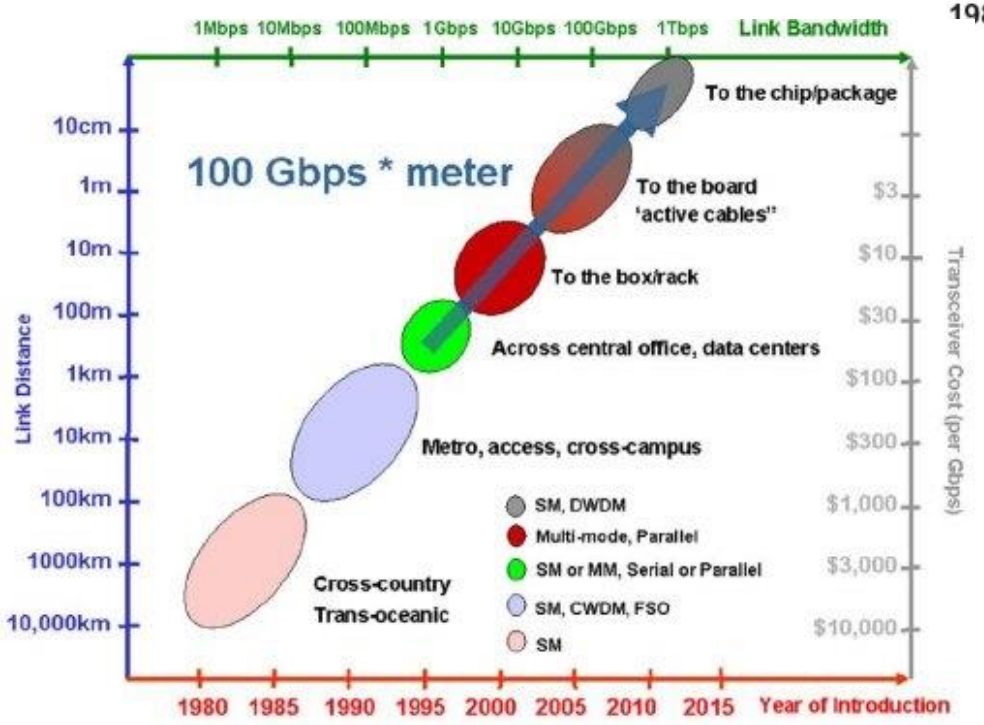
+Testing !

How – MPW runs and Photonic Ecosystem

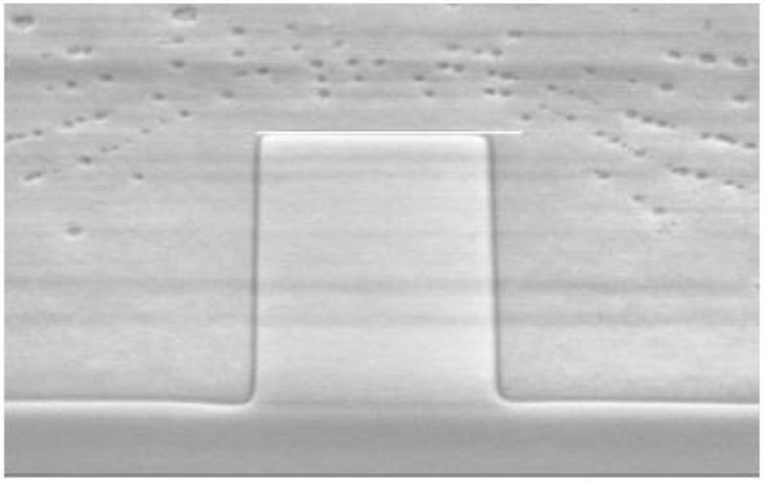
The European photonic ecosystem




Transmission Capacity/fiber (bps)
1 E
100 P
10 P
1 P
100 T
10 T
1T
100 G
10 G
1 G
100 M
New Multiplexing: SDM/~MDM
Pbit/s~Ebit/s
New Fibers
Amplifier Bandwidth
SMF Input Limit
Optical Power Limit (Nonlinearities, Fiber fuse)
Digital Coherent
EDFA, WDM
1.5dB/year
TDM
LD+Fiber
1980
1990
2000
2010
2020
2030
2040
Year

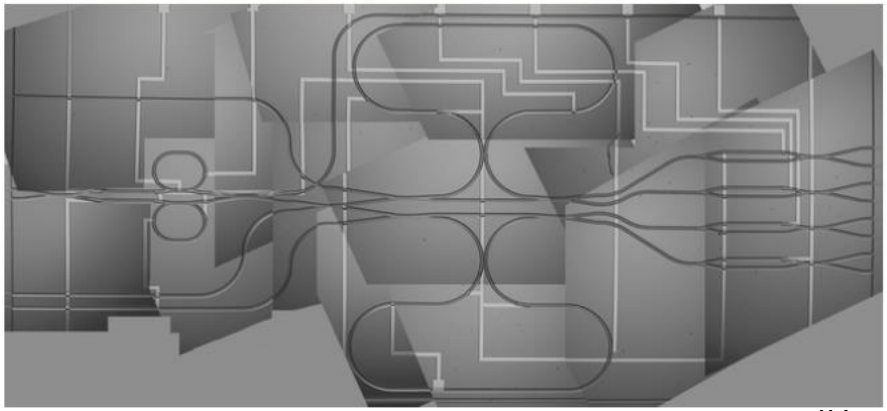

1Mbps
10Mbps
100Mbps
1Gbps
10Gbps
100Gbps
1Tbps
Link Bandwidth
100 Gbps * meter
To the chip/package
To the board 'active cables"
To the box/rack
Across central office, data centers
Metro, access, cross-campus
Cross-country Trans-oceanic
SM, DWDM
Multi-mode, Parallel
SM or MM, Serial or Parallel
SM, CWDM, FSO
SM
10cm
1m
10m
100m
1km
10km
100km
1000km
10,000km
Link Distance
$3
$10
$30
$100
$300
$1,000
$3,000
$10,000
Transceiver Cost (per Gbps)
1980
1985
1990
1995
2000
2005
2010
2015
Year of Introduction

Glass on Silicon



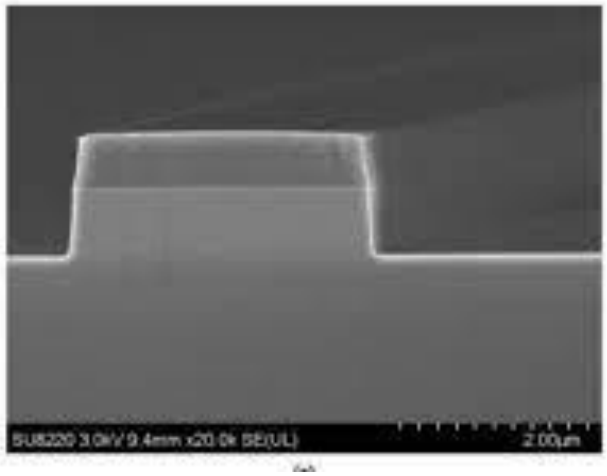
Ge:SiO_2, SiON, SiOC

Dielectric, amorphous, n=1.45…1.6
Size: w= 3…6 μm
Transparency: vis …. 3 μm
Passive devices
Low integration scale (R= 300 μm … 5mm)
Excellent coupling with Optical fibers
Suitable for mid volume production
Very Low Losses (0.1 dB/cm)

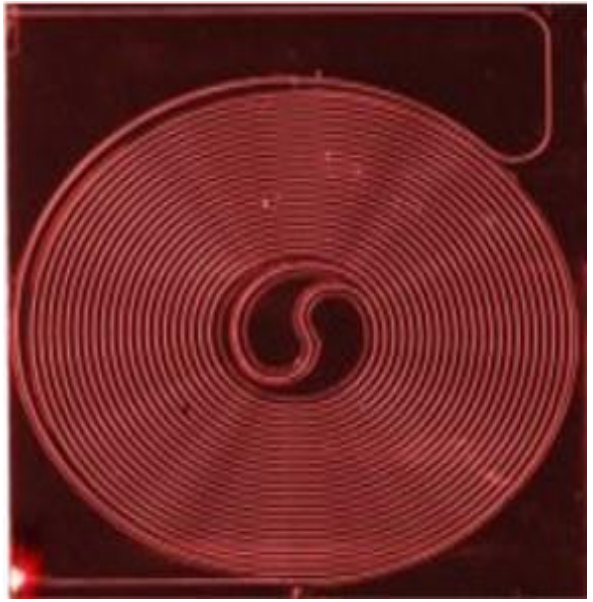


Silicon Nitride SiN

Si_3N_4, SiRN

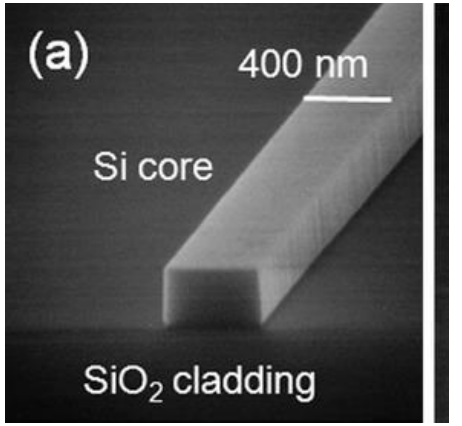


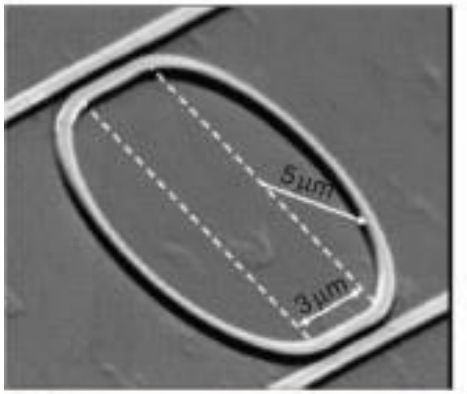
Dielectric, amorphous, n=1.99
Size: w= 1…5 μm, t=50…800 nm
Transparency: vis …. 4 μm
Passive devices
High integration scale (R= 20 μm)
Heterogeneous integration possible
Good coupling with Optical fibers
Suitable for mid volume production
Very Low Losses (0.1 dB/cm, record 0.1 dB/m)



POLITECNICO DI MILANO

Silicon on Insulator (SOI)





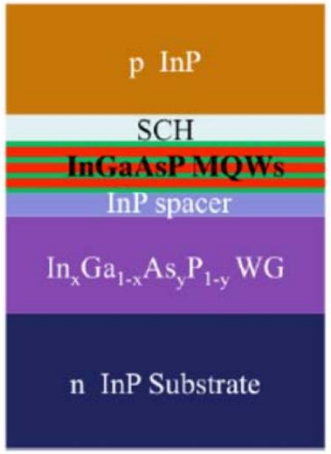


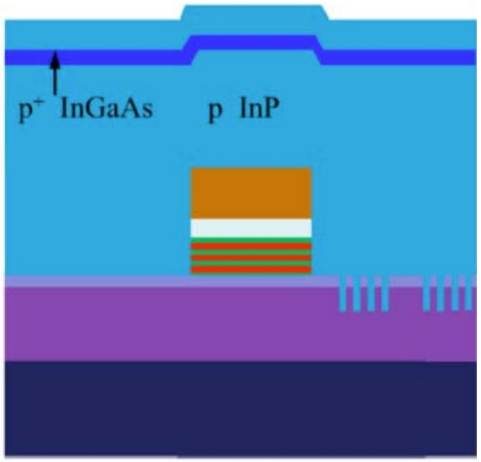
- Semiconductor, crystal, n=3.45
- Size: w= 0.5 μm, t=220 nm
- Transparency: 1.1…. 2.5 (10) μm
- Passive devices
- p-n junctions (modulators)
- Ge photodetector
- Very high integration scale (R= 5μm)
- Monolithical integration with electronics (CMOS)
- Difficult coupling with Optical fibers
- Suitable for mass production
- ”High Losses” (1dB/cm)

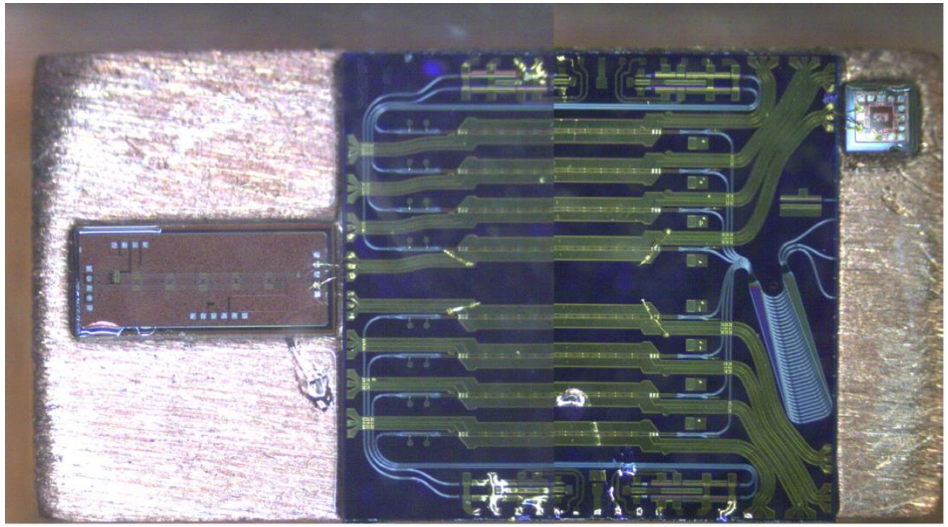
Indium Phosphide

InP

Semiconductor, Crystal, n=3.0….3.4
Epitaxial grow, quantum well
Complex technology
Size: “2x2” μm
Transparency: 1.2…1.6 μm
Active devices, modulators, also passive devices
Mid integration scale (R= 100 μm)
Heterogeneous integration difficult but possible
Good coupling with Optical fibers
Suitable for mid volume production
Losses 0.5-3 dB/cm







Lithium Niobate

$LiNbO_3$

Electrooptic Crystal, n=2.2
Diffused; Thin film
Size: 10x10 μm; w= 1…2 μm, t=300…600 nm
Transparency: vis …. 4 μm
Passive devices, modulators
Mid integration scale (R= 50 μm, diffused 5cm)
Heterogeneous integration difficult
Lossy coupling with Optical fibers
Suitable for mid volume production (?)
Losses 0.5 dB/cm

